PETRO-QUANT 4.1 for S8 TIGER Series 3

User Manual

Innovation with Integrity

Bruker AXS - XRF

Table of Contents

1 Introduction

PETRO-QUANT is the Bruker AXS solution for the easy analysis of trace elements, wear metals, and additives in all kinds of petrochemical materials. PETRO-QUANT 4.1 combined with the S8 TIGER III wavelength dispersive XRF spectrometers offers unrivalled accuracy. It is a ready-to-analyze package for multi-elemental analysis in various petrochemical or other light matrices right out of the box.

PETRO-QUANT 4.1 BASIC (successor of PETRO-QUANT 4 BASIC) is the Bruker AXS universal calibration for trace analysis in samples of petrochemical matrix (H to F > 95 %) for the S8 Tiger Series III. S and Cl can be present in the range from lowest mg/kg level up to 5 %. The additive elements Mg, P, Ca, Zn, and Ba are expected to be in a range up to several tenths of a percent. All other 25 elements are in the trace concentration range (less than 1000 mg/kg). The calibration and precision parameters are presented in chapter *What Will PETRO-QUANT 4.1 Do for You? [▸ 4]*

1.1 What Will PETRO-QUANT 4.1 Do for You?

PETRO-QUANT 4.1 is pre-configured to allow “PETRO-analysts” to begin sample measurements quickly and easily. Furthermore, the application long-term maintenance (quality control measurements, drift correction) is reduced to a minimum workload for a routine lab.

The PETRO-QUANT 4.1 BASIC method provides:

- Analytical lines carefully designed to:
  - Use optimum tube and instrument settings.
  - Collect peak and background intensities with a minimum of line overlaps.
  - Use carefully selected Pulse Height Discriminator windows.
- Counting strategy to yield best accuracy within 19 minutes (S8 TIGER Series 3) measuring time
- Long term Drift Correction by use of well established oil samples
- Instructions for easy and reliable preparation methods to produce safe liquid samples
- Line overlap, background and matrix corrections are automatically applied
- Flexibility for modification of counting times and the usage of PETRO-QUANT <Be> lines in user defined calibration methods
- Automatic correction for non-infinitely thick samples (sample height lower than critical depth)
- Automatic selection of best line
- Automatic determination of oxygen contents

The PETRO-QUANT 4.1 BASIC method covers the following list of analytes and respective concentration ranges:

	Concentration-Range [mg/kg]	**Regression-Deviation [mg/kg]**	**LoD (3σ; 12/4 s) [mg/kg]**	**LoD (3σ; 100/100 s) [mg/kg]**	**Avg.Abs.Dev. | Range (~100 mg/kg; n=12) [mg/kg]**
Na	LoD - 2000	5.7	3.2**	1.5	2.3 | 8
Mg	LoD - 4000	7.0	2.7**	1.3	0.8 | 3

	Concentration-Range [mg/kg]	Regression-Deviation [mg/kg]	LoD (3σ; 12/4 s) [mg/kg]	LoD (3σ; 100/100 s) [mg/kg]	Avg.Abs.Dev. | Range (~100 mg/kg; n=12) [mg/kg]
Al	LoD - 1250	2.7	2.4	0.6	1.2 | 5
Si	LoD - 2000	3.4	2.7	0.7	0.6 | 2
P	LoD - 5000	3.8	0.6	0.1	0.5 | 2
S	LoD - 1000	10	0.7	0.2	0.9 | 3
S*	LoD – 2.50 %	0.0049 %	~6	~1	
Cl	LoD - 1000	5.9	2.7	0.7	1.2 | 4
Cl*	LoD – 2.00 %	0.0014 %	~7	~2	
K	LoD - 2000	4.5	1.1	0.3	0.3 | 2
Ca	LoD - 5000	20.6	1.1	0.3	0.6 | 2
Ti	LoD - 600	0.9	0.8	0.2	0.4 | 2
V	LoD - 600	1.0	0.6	0.1	0.2 | 1
Cr	LoD - 600	0.8	0.6	0.1	0.4 | 1
Mn	LoD - 600	0.9	0.6	0.2	0.3 | 1
In	LoD - 600	2.2	0.5	0.2	1.0 | 6
Sn	LoD - 600	2.2	2.3	0.6	1.0 | 4
Sb	LoD - 600	1.5	2.5	0.6	0.8 | 4
Ba	LoD - 8000	9.2	2.6	0.6	0.6 | 2
Fe	LoD - 2000	5.3	0.5	0.1	0.3 | 2
Co	LoD - 600	0.9	0.3	0.1	0.8 | 5
Ni	LoD - 600	2.1	0.4	0.1	0.4 | 2
Cu	LoD - 600	3.2	0.4	0.1	0.5 | 2
Zn	LoD - 2000	6.1	0.3	0.1	0.3 | 1
As Kα As Kβ	LoD – 182	2.0 1.7	0.2 1.0	0.1 0.3	0.4 | 2*** 0.7 | 4***
Br	LoD - 1200	7.5	0.2	0.1	0.6 | 2
Zr	LoD - 600	1.9	0.6	0.1	0.3 | 1
Mo	LoD - 1250	2.5	0.7	0.2	0.5 | 2
Ag	LoD - 600	2.4	2.7	0.7	1.6 | 7
Cd	LoD - 600	2.1	2.4	0.6	1.1 | 5
W Lα W Lβ	LoD – 1250	2.6 2.6	0.6 0.6	0.1 0.1	0.7 | 5 0.5 | 3
Tl Lα Tl Lβ	LoD – 600	1.8 2.4	0.3 0.7	0.1 0.2	0.5 | 4 0.6 | 4

	Concentration-Range [mg/kg]	Regression-Deviation [mg/kg]	LoD (3σ; 12/4 s) [mg/kg]	LoD (3σ; 100/100 s) [mg/kg]	Avg.Abs.Dev. | Range (~100 mg/kg; n=12) [mg/kg]
Pb Lα Pb Lβ	LoD - 1250	4.3 2.1	0.4 0.8	0.1 0.2	0.5 | 2 0.5 | 3
Bi Lα Bi Lβ	LoD - 600	2.2 5.3	0.5 1.0	0.1 0.3	0.5 | 3 0.8 | 3

*S and Cl have one line for trace concentrations and one line for minor concentrations

**32/16s measurement time

***For As, repeatability was determined at 75 mg/kg concentration

Table 1.1: Concentration ranges and analytical limits for PETRO-QUANT (Scintillation counter systems)

	Concentration-Range [mg/kg]	Regression-Deviation [mg/kg]	LoD (3σ; 12/4 s) [mg/kg]	LoD (3σ; 100/100 s) [mg/kg]	Avg.Abs.Dev. | Range (~100 mg/kg; n=12) [mg/kg]
Na	LoD - 2000	8.9	4.3**	1.5	3.6 | 15
Mg	LoD - 4000	5.6	3.0**	1.3	1.1 | 6
Al	LoD - 1250	4.5	2.6	0.6	1.7 | 6
Si	LoD - 2000	2.1	2.8	0.7	0.6 | 3
P	LoD - 5000	3.5	0.6	0.1	0.3 | 1
S	LoD - 1000	8	0.8	0.2	0.6 | 3
S*	LoD – 2.50 %	0.0057 %	~6	~1	
Cl	LoD - 1000	4.7	2.9	0.7	1.3 | 6
Cl*	LoD – 2.00 %	0.0015 %	~7	~2	
K	LoD - 2000	2.9	1.1	0.3	0.5 | 3
Ca	LoD - 5000	24.9	1.1	0.3	0.6 | 2
Ti	LoD - 600	0.8	0.9	0.2	0.3 | 1
V	LoD - 600	0.6	0.7	0.2	0.4 | 2
Cr	LoD - 600	0.5	0.7	0.2	0.5 | 2
Mn	LoD - 600	0.7	0.6	0.2	0.2 | 1
In	LoD - 600	0.9	2.5	0.6	1.1 | 4
Sn	LoD - 600	1.4	2.3	0.6	0.9 | 1
Sb	LoD - 600	1.2	2.5	0.6	0.8 | 3
Ba	LoD - 8000	8.2	2.6	0.6	1.5 | 5
Fe	LoD - 2000	2.5	0.5	0.2	0.7 | 3
Co	LoD - 600	0.7	0.2	0.1	0.5 | 2

	Concentration-Range [mg/kg]	Regression-Deviation [mg/kg]	LoD (3σ; 12/4 s) [mg/kg]	LoD (3σ; 100/100 s) [mg/kg]	Avg.Abs.Dev. | Range (~100 mg/kg; n=12) [mg/kg]
Ni	LoD - 600	0.9	0.3	0.1	0.3 | 1
Cu	LoD - 600	0.9	0.4	0.1	0.3 | 1
Zn	LoD - 2000	1.8	0.4	0.1	0.3 | 2
As Kα As Kβ	LoD – 182	3.6 2.2	0.2 1.5	0.1 0.4	0.3 | 1 1.0 | 4
Br	LoD - 1200	1.9	0.2	0.1	0.5 | 2
Zr	LoD - 600	0.5	0.7	0.2	0.5 | 3
Mo	LoD - 1250	1.5	1.0	0.2	0.4 | 1
Ag	LoD - 600	2.1	3.4	0.8	2.0 | 8
Cd	LoD - 600	1.2	3.4	0.8	1.3 | 6
W Lα W Lβ	LoD – 1250	0.7 1.2	0.5 1.2	0.1 0.3	0.7 | 3 0.8 | 4
Tl Lα Tl Lβ	LoD – 600	1.8 4.7	0.3 0.7	0.1 0.2	0.5 | 2 0.6 | 3
Pb Lα Pb Lβ	LoD - 1250	3.4 2.9	0.4 0.7	0.1 0.2	0.6 | 3 0.3 | 1
Bi Lα Bi Lβ	LoD - 600	1.2 1.0	0.9 0.5	0.2 0.1	0.4 | 2 0.4 | 2
*S and Cl have one line for trace concentrations and one line for minor concentrations **32/16s measurement time ***For As, repeatability was determined at 75 mg/kg concentration					

Table 1.2: Concentration ranges and analytical limits for PETRO-QUANT (HighSense systems)

The default measurement method including peak and background positions produces data for all of these analytes within 19 minutes measuring time.

1.2 What Will PETRO-QUANT 4.1 NOT Do?

PETRO-QUANT 4.1 is designed to analyze traces, minors and majors in a wide variety of petrochemical matrices. It is not intended to analyze:

- Samples that have significant amounts of elements not listed in the two tables above (see chapter). *Using PETRO-QUANT Lines in Custom Methods [▸ 38]*)
- Samples that have concentration ranges clearly above those listed in the two tables above (e.g. additive concentrates; but see chapter *Using PETRO-QUANT Lines in Custom Methods [▸ 38]*).
- X-ray fluorescence is a comparative technique. PETRO-QUANT 4.1 is based on homogeneous samples with a flat film at the bottom of the liquid cup (no sagging). To aim for best performance you should ensure that your liquid samples fulfill these

requirements. Three procedures (outlined in chapter *PETRO-QUANT 4.1 Sample Preparation [▶ 23]*) and their preparation materials are provided to help you in creating good and safe liquid samples.

- When analyzing samples using PETRO-QUANT 4.1, it is important to note that the software, method and spectrometer are not able to correct for the following sample effects:
  - Non-homogeneous samples
  - Segregating wear metals
  - Sagging sample cups films

The PETRO-QUANT 4.1 BASIC measurement parameters are not designed for the lowest Limit of Detection (LoD) and minimum measurement times for the analysis of one specific sample type. Measurement methods and parameters are optimized for a universal multi-elemental analysis system for all kinds of petrochemical samples. Therefore, the measurement time per element as well as number and position of background positions can be optimized for specific sample types (see chapter *Customizing Measurement Methods and Applications [▶ 30]*).

See also

Reusing PETRO-QUANT Lines for Custom Calibrations [▶ 39]

1.3 If You Have the SampleCare Option PETRO-QUANT Be

If your S8 Tiger Series 3 is configured with a 50 µm Be filter for SampleCare, your PETRO-QUANT will come with two versions of the solution installed: The regular PETRO-QUANT, and the Be filter version PETRO-QUANT Be, which uses the Be filter for every line where no filter is otherwise applied.

The two versions of the solution are installed on your database as separate methods and have to be drift corrected and validated separately. The initial drift and validation procedure should be carried out on both solutions (see chapter *PETRO-QUANT 4.1 Installation and Maintenance [▶ 11]*). If you only use one of the methods regularly, you can drift and blank correct as needed.

The use of the Be filter causes increased lower limits of detections for the light elements due to partial absorption of the exciting radiation, but it can protect the X-ray tube in the case of more aggressive sample types that come with a higher risk of foil degradation. The LLD increase is especially pronounced for Na (~30%) and Mg (~15%). For the other light elements up to Ca, an increase between 5 and 10% can be expected.

2 X-ray Fluorescence Analysis of Liquid Samples

When analyzing liquid samples using XRF, there are two important considerations:

- The measurements of liquid samples cannot be performed in vacuum mode. Helium mode with reduced (samples of low volatility) or atmospheric pressure (samples of high volatility) is used.
- Liquids are analyzed in liquid cups. These are made of two polymer rings and a polymer lid. The bottom of these cups is made of a very thin film.

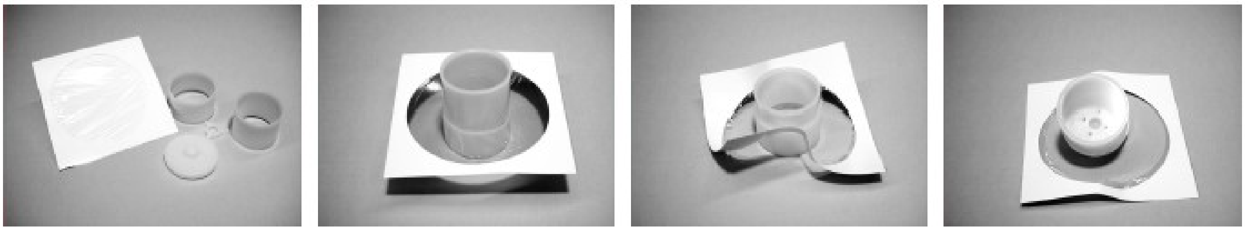

Figure 2.1: Figure 2.1: Preparation of liquid cups

2.1 Measurement in He atmosphere

Using a S8 TIGER Series 3 spectrometer, two Helium modes are available in SPECTRA.ELEMENTS:

- Helium with reduced pressure of about 150-250 mbar ("Helium (reduced pressure)")
- Helium with atmospheric pressure ("Helium (atmospheric pressure)")

The mode is defined in the measurement method in SPECTRA.ELEMENTS depending on the vapor pressure of the sample. The atmospheric pressure He mode allows the analysis of highly volatile samples like gasoline. In contrast the reduced He mode consumes less Helium and allows the analysis of low volatile samples like oils. If PETRO-QUANT 4.1 standard methods are used, the measurement parameters are already set and the correct mode is selected via the choice of application (see *PETRO-QUANT 4.1 Routine Analysis [▶ 24]*).

NOTICE
The measurement chamber is evacuated in standby mode. If an error message occurs during measurement (for example, if the EasyLoader loses track of the sample during transfer), always remove any liquid samples from the measurement chamber before acknowledging the error message on the touch screen or external computer!

2.2 Sample Preparation in Liquid Cups

The choice of the film material for the liquid cup bottom is essential for the analysis of liquids (and loose powders).

In general, all elements from Na to U can be analyzed in liquids independent from the film material and consequently, the absorption of the fluorescence lines. Unfortunately, the fluorescence lines of lighter elements do not penetrate the film. In these instances, the choice of the film material is always a compromise between low absorption and on the other hand mechanical and chemical stability.

Proven film materials for XRF liquid analysis are:

- 4 µm Prolene® (Prolene is a registered trademark of Chemplex Industries Inc.) film: most often used transparent film and free of contaminations; material of choice if not chemically attacked (e.g. by gasoline samples).
- 3.6 µm Mylar® (Mylar is a registered trademark of DuPont de Nemours Inc.) film: mechanically more stable than Prolene® film, being less transparent and contaminated by P, Ca and Si (see chapter *Compensation of Film Materials [▸ 10]*); material of choice for gasoline samples.

NOTICE

The appropriate film for unknown liquids can be tested easily. Each sample is poured into two cups being prepared with the different film materials and placed on a paper tissue as seepage warning device. If the Prolene® film is immediately attacked and crinkles (e.g. by gasoline) you can change to Mylar® film. If after hours the liquid cup is lifted and neither the tested film sags nor the paper tissue indicates any stains, experience shows the film material withstands XRF measurements of up to at least 30 min.

2.3 Compensation of Film Materials

Using one universal calibration like PETRO-QUANT 4.1<Be> BASIC for a big variety of samples requires the application of more than one preparation procedure. To adjust the calibration offset and to compensate film contaminations for different preparations, PETRO-QUANT 4.1 uses a blank sample.

Please note, using contaminated film materials without a blank correction results in incorrect concentrations of the respective elements. Due to differences in sample composition, the effect of these film contaminations on the final matrix corrected concentrations can also vary from sample to sample. The 3.6 µm Mylar® film is contaminated by traces of P, Ca and Si. The corresponding preparation is "PETRO-QUANT 4.1<Be>-Fuel”. The compensation of the film contamination must always be based on a blank measurement executed with the same film.

A detailed description of how to measure the blank sample can be found in chapter *How to Perform a Blank Correction [▸ 14]*. When you measure a blank sample according to the instructions, the blank correction of the calibration and final concentrations is done automatically by the SPECTRA.ELEMENTS software. If a custom preparation is defined with a different film, the same blank sample can be used, but the preparation procedure (esp. film material) should match that which is used for real measurements.

This blank measurement should be done anew with any new lot (roll or box) of film material, even when keeping the same application and preparation method.

3 PETRO-QUANT 4.1 Installation and Maintenance

The PETRO-QUANT 4.1 BASIC method is installed, tested and validated on your S8 TIGER Series 3 before it leaves the factory. All the associated data and measurements are stored in the instrument database. The following steps have to be completed for validation after the instrument has been installed in your lab:

1. Drift and blank correction: The drift correction compensates for any small changes in intensity that may have occurred following transportation and installation of your system. The blank correction takes care of signal due to impurities in the foil, and should be done with the same batch of foil that you use to measure your QC and Unknown samples. To carry out the drift correction, the blank and drift samples have to be measured. A detailed description can be found in chapter *Drift Correcting PETRO-QUANT 4.1 [▸ 12]*. After this step, the PETRO-QUANT 4.1 "Oil" method is ready to measure your petrochemical sample (however, step 2 (quality control) is recommended before measuring unknown samples).
2. Quality control: In order to ensure good data quality and validate the drift corrected method, quality control (QC) samples are provided. These samples have known values for all PETRO-QUANT 4.1 analytes and are used as check samples to validate the installation of PETRO-QUANT 4.1 on your S8 TIGER Series 3. QC measurements should also be periodically carried out to check the validity of the method to determine when an additional drift correction becomes necessary.
3. *Optional:* "Fuel" Blank correction: If you want to use the "Fuel" method (atmospheric He, Mylar), a Blank sample triplicate should be measured for this method using the 3.6 µm Mylar® film. If you are only using the "Oil" method, this step is not required.

The following table briefly summarizes the recommended measurement steps directly after installation of the Tiger III in your laboratory:

	Loader tab	**Sample**	**Mode/Method**	**Preparation**	**Detailed instruction**
1	Blank Standards*	BLANK (3x)	Helium (reduced pressure) / DC Oil	7.0 g, 4 µm Prolene®	*How to Perform a Blank Correction [▸ 14]*
2	Drift Specimens	DC Con S21&K&Sb 500 ppm (2x)	Helium (reduced pressure) / DC Oil	7.0 g, 4 µm Prolene®	*How to Perform a Drift Correction (Sensitivity) [▸ 12]*
3	Control Specimens	QC Paraffin	Helium (reduced pressure) / Oil		*How to Validate PETRO-QUANT 4.1 [▸ 18]*
		QC Con S21+K+Sb 100ppm			
		QC-XRS1			
		QC-XRS2			
4	*Blank Standards**	*BLANK (3x)*	*Helium (atmospheric pressure) / DC Fuel*	*7.0 g, 3.6 µm Mylar®*	*How to Perform a Blank Correction [▸ 14]*

Table 3.1: Measurement steps after on-site installation to correct for drift during delivery and validate PETRO-QUANT.

*The Blank standards should be measured with the same batch of foil that is used for the measurement of Control and Unknown samples.

3.1 Drift Correcting PETRO-QUANT 4.1

The full drift correction is carried out once after installation of the instrument in your laboratory, and then over time based on the results of the measurements of the respective QC samples (see *Validating PETRO-QUANT 4.1 [▸ 17]*). Additional blank corrections may be needed more frequently, for example when the foil batch used for sample preparation is changed.

The PETRO-QUANT 4.1 drift correction strategy is the following:

The PETRO-QUANT 4.1 BASIC methods are drift corrected on the basis of oil samples included in delivery. The outstanding stability of the S8 TIGER Series 3 spectrometers ensures that the PETRO-QUANT 4.1 drift correction should not be necessary for weeks to months. In general, there are two different tasks the PETRO-QUANT 4.1 drift correction procedure takes care of:

The first one is the compensation of the instrument´s sensitivity drift. This is the classical drift correction that compensates the aging effects of the instrument`s tube and detectors and is based on a reference sample with known concentrations of the reference elements (DC sample, see chapter *How to Perform a Drift Correction (Sensitivity) [▸ 12]*).

The second one is the compensation of scattered radiation of film contaminants and suboptimal background modeling of a given sample type and preparation, i.e. subtraction of a blank sample's intensity. The blank correction uses the “BLANK” sample (comprised of pure paraffin) and hence compensates changes in the offsets of the respective calibrations (see chapter *How to Perform a Blank Correction [▸ 14]*).

3.1.1 How to Perform a Drift Correction (Sensitivity)

The PETRO-QUANT 4.1 drift correction is based on the drift correction reference samples which are included in delivery. Drift corrections for every measurement mode (“DC Oil” and “DC Fuel”) are performed during the factory installation of PETRO-QUANT. The effect of the measurement method (Helium atmosphere) on the intensity is therefore calculated and stored in the database. Thus, the drift correction only has to be performed with one method (usually DC Oil) to correct for the changes over time, since drift data for the other application is then automatically calculated from this.

To start a drift measurement:

1. Select the “Drift Specimen” tab in the LOADER in your program or on your touchscreen. Under “Options”, select the pressure mode (usually Helium (reduced pressure)) and method (usually “DC Oil”) under “Options”. The drift samples (duplicates) automatically appear once the method is selected (see image below).
2. Prepare each sample individually in a liquid cup according to the table below. Place cup 30s on a non-contaminating underlay for leakage testing, put liquid cup into sample holder with 34 mm aperture, place it on the respective Loader position and start measurement.

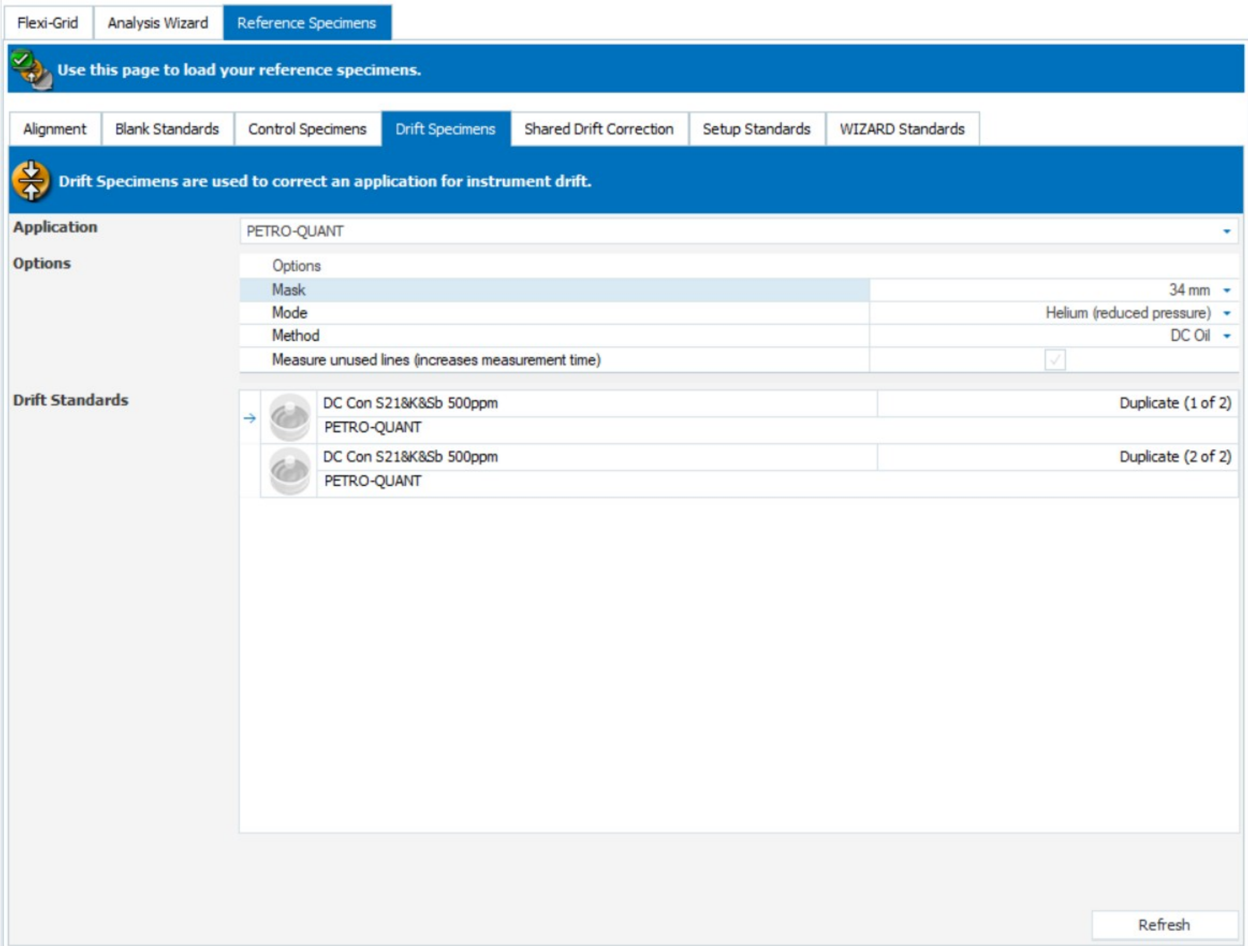

Figure 3.1: Drift Samples in the LOADER.

The preparations and settings that should be used during the drift correction for the respective measurement method are summarized in the following table:

Application to be corrected	Drift sample	Atmosphere, Drift method	Preparation method
PETRO-QUANT 4.1 Oil	DC Con S21&K&Sb 500 ppm	Helium (reduced pressure), DC Oil	7.0 g, 4 µm Prolene®
PETRO-QUANT 4.1 Fuel	DC Con S21&K&Sb 500 ppm	Helium (atmospheric pressure), DC Fuel	7.0 g, 4 µm Prolene®

Table 3.2: Drift correction settings and sample preparations

As mentioned above, correction factors between the different measurement methods are stored in the database, and thus the drift correction has to be regularly performed for only one of the two measurement methods to correct both applications for instrument drift over time. However, if you perform a drift correction using “DC Fuel”, it is important to note that the preparation should be kept the same as the calibration preparation (using the 4 µm Prolene® foil).

NOTICE

Measure Unused Lines

In PETRO-QUANT, the “Measure Unused Lines” option is on by default for drift and blank measurements. This is the recommended setting since it preserves the original measurement order at calibration time, resulting in the most accurate intensities.
It is possible to change this setting in the main solution node of PETRO-QUANT in the WIZARD plugin. Deselecting this option reduces the drift measurement time by around 6 minutes, but can lead to small systematic errors (between 0.5-1.5% relative error) in the Compton line and for light elements (Na-P).

3.1.2 How to Perform a Blank Correction

All PETRO-QUANT 4.1 methods are blank corrected based on a pure paraffin sample included in the delivery. The blank correction principle for PETRO-QUANT is based on the measurement method: For all “Oil” methods, which are measured under Helium (reduced) atmosphere, the Prolene® foil is used. Thus, the Blank sample measured with “DC Oil” should be prepared using the Prolene® foil. For the “Fuel” application, the Mylar® foil is used, therefore the Blank samples measured under “DC Fuel” should be prepared using the Mylar® foil. For both applications, separate Blank samples have to be measured by choosing the correct application (“DC Oil” or “DC Fuel”) in the LOADER. Only applications that have a valid blank sample measured with the corresponding drift method can be used to evaluate unknown samples. It is recommended to perform a new blank measurement for every new batch of foil.

To measure a blank sample:

1. Select the “Blank Standards” tab in the SPECTRA.ELEMENTS LOADER. Select the atmosphere and application you would like to blank correct (“Helium (reduced pressure)” and “DC Oil” or “Helium (atmospheric)” and “DC Fuel”) under “Options”. The BLANK samples (triplicates) automatically appear (see image below). Check the “measure unused lines” option.
2. Prepare each sample individually according to the table below. Place cup 30s on a non-contaminating underlay for leakage testing, put liquid cup into sample holder with 34 mm aperture, place it on the respective Loader position and start measurement.

NOTICE

Handling of duplicate samples

If you measure duplicates (or multiples of any number) in SPECTRA.ELEMENTS, the samples are usually placed into the LOADER and selected together. However, it is advisable to measure only one freshly prepared sample at a time. Before starting the measurement, make sure that only the one duplicate you would like to measure is selected, since clicking on duplicate usually selects all of them simultaneously.

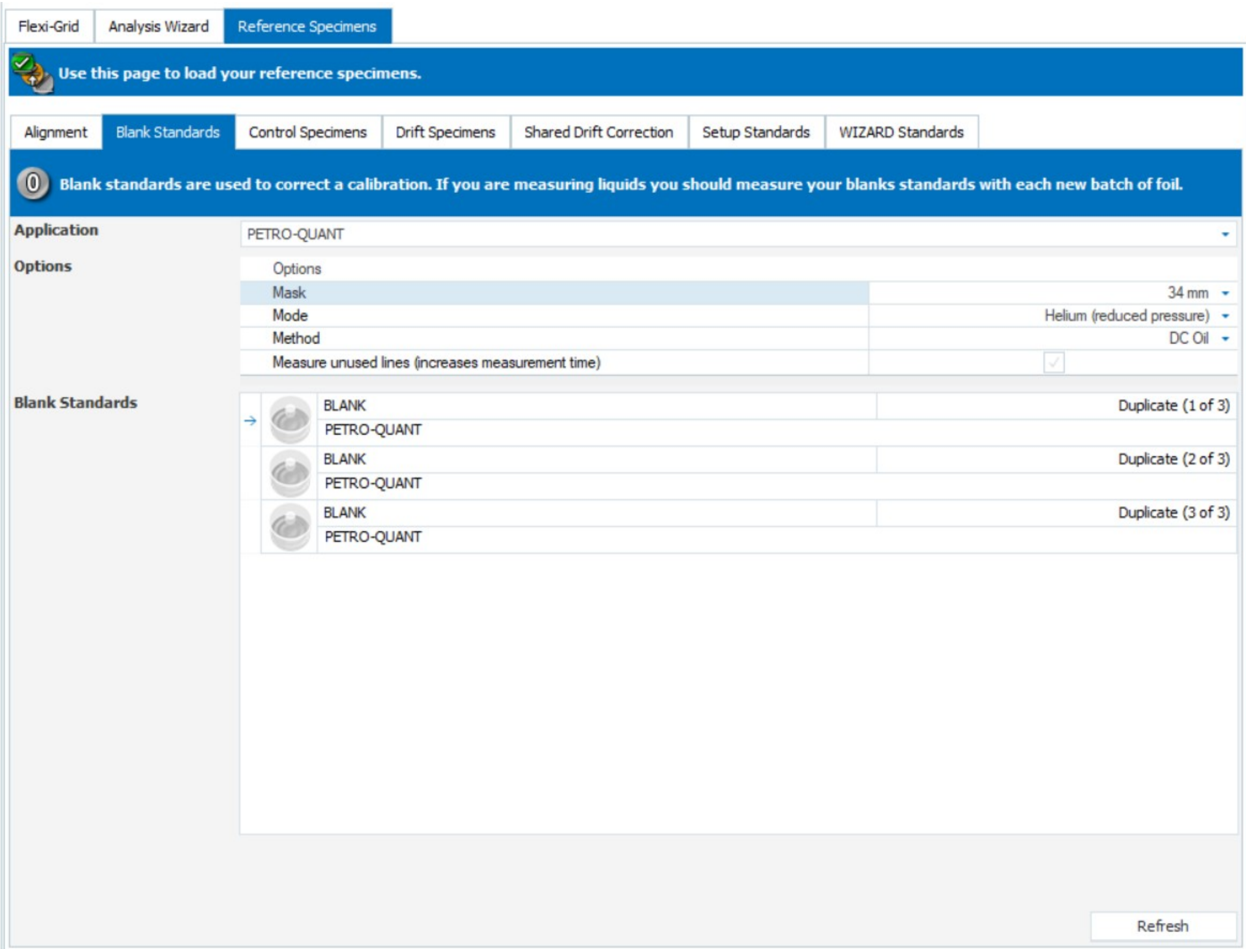

Figure 3.2: Blank samples in the LOADER

The currently used Blank measurement of the method can be viewed by opening the method in the WIZARD of the SPECTRA.ELEMENTS software under the “Calibrated Materials” section by clicking on “Show Blanks”. The most recent blank measurement measured before the respective unknown sample that matches the measurement method (Helium reduced or atmospheric) is always used for evaluation.

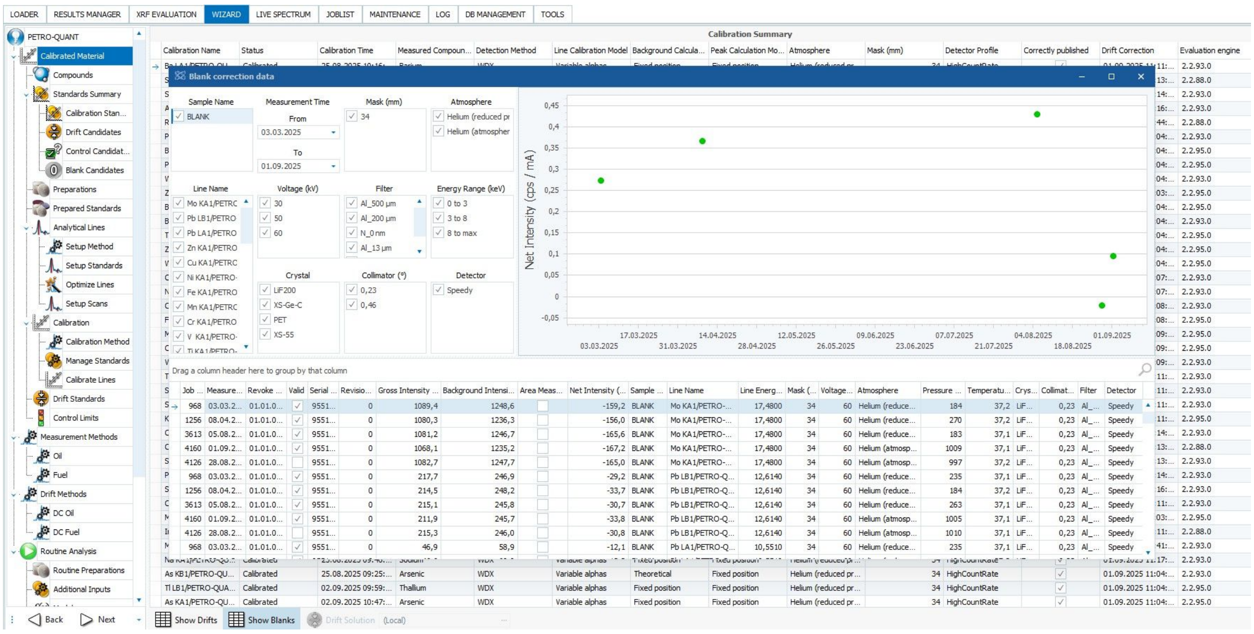

Figure 3.3: Existing blank data depicted in the SPECTRA.ELEMENTS WIZARD. The latest blank measurement with matching measurement method is used for correction of future measurements.

Application to be corrected	Blank sample	Measurement method	Preparation method
PETRO-QUANT 4.1 Oil	BLANK (3x)	DC Oil	7 g, 4 µm Prolene®
PETRO-QUANT 4.1 Fuel	BLANK (3x)	DC Fuel	7 g, 3.6 µm Mylar®

Table 3.3: Blank preparations and measurement methods

NOTICE

Measure unused lines

In PETRO-QUANT, the “Measure Unused Lines” option is on by default for drift and blank measurements. This is the recommended setting since it preserves the original measurement order at calibration time, resulting in the most accurate intensities. It is possible to change this setting in the main solution node of PETRO-QUANT in the WIZARD plugin. Deselecting this option reduces the drift measurement time by around 6 minutes, but can lead to small systematic errors (between 0.5-1.5% relative error) in the Compton line and for light elements (Na-P).

3.1.3 Drift Correcting PETRO-QUANT Be

If your S8 Tiger Series 3 comes with a Be filter option, your database will contain the SampleCare version of the solution, PETRO-QUANT Be, alongside the standard PETRO-QUANT (where no filter is used for many of the lines). The first drift and blank correction after installation of the system in your laboratory should always be carried out for both solutions in order to ensure the solutions can be used. Afterwards, regular drift correction should be carried out when you are using the solution.

The blank and drift correction procedure for PETRO-QUANT Be is exactly the same as that for the regular PETRO-QUANT. In order to perform the correction for PETRO-QUANT Be, repeat the steps described in chapters *How to Perform a Drift Correction (Sensitivity) [▸ 12]*

and *How to Perform a Blank Correction [▷ 14]*, but select “PETRO-QUANT Be” as the solution in the LOADER. The same combination of measurement methods and sample preparations should be used.

3.2 Validating PETRO-QUANT 4.1

The validation of PETRO-QUANT 4.1 is carried out using the Quality Control (QC) samples. A validation should be measured directly after the first drift correction after installation of the instrument in your laboratory, and then periodically to confirm the validity of the method over time and check for potential instrument drift.

The PETRO-QUANT 4.1 Analytical Acceptance Test documents provide an Acceptance Certificate which includes tables showing consensus values, allowed tolerances, factory derived results and a space to enter your measured values collected from PETRO-QUANT 4.1 methods. The tolerance values allow for analytical error, preparation error and the reported uncertainty of the consensus values. The respective tables need completing, checking and validating. If the data is within the allowed tolerances then the installation has been successful, and the Acceptance Certificates can be signed and stored with the system documentation.

NOTICE

Please note that the values used in the acceptance procedure are ‘consensus’ values. They are presented by the issuing body with a range of estimated error and notes of caution regarding accuracy. The analytical tolerances quoted in the acceptance procedure take into account the estimated errors of the standard values, analytical error and sample preparation error.

3.2.1 Performing an S8-Check Prior to Validating PETRO-QUANT

Before performing the quality control measurement for PETRO QUANT, please check the instrument by using the S8-Check method.

We generally recommend to schedule the measurement of `S8-Check` once a day. Refer to the user manual of S8 Tiger III to know more.

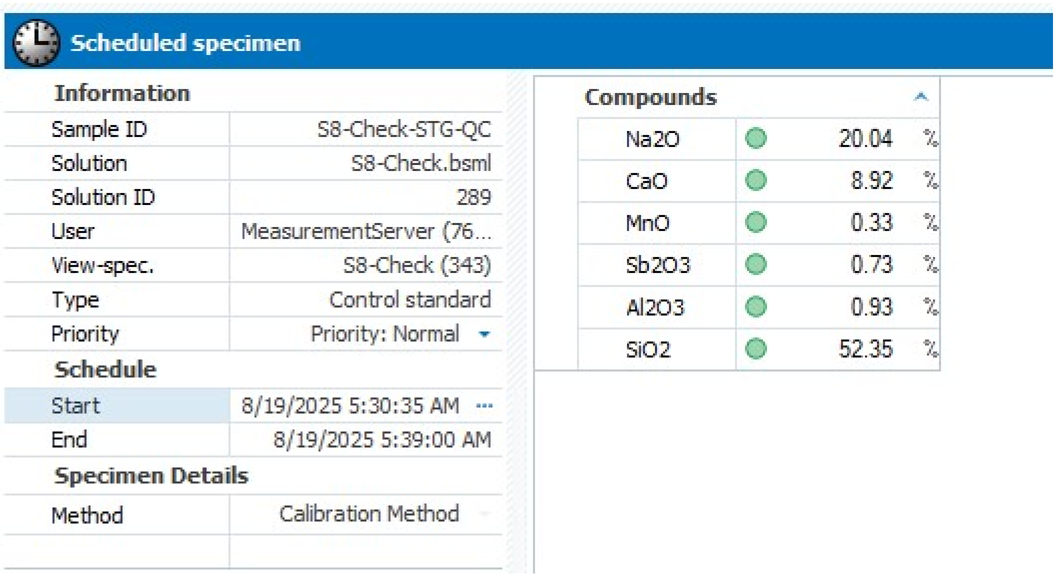

Figure 3.4: S8-Check

3.2.2 How to Validate PETRO-QUANT 4.1

The quality control standards provided with PETRO-QUANT 4.1 BASIC are listed below, together with the required measurement settings:

Standard	Application	Preparation
QC Paraffin	PETRO-QUANT 4.1 - Oil	Sample mass 7.0±0.1g Liquid cup 4 µm Prolene®
QC Con S21+K+Sb 100ppm		
QC-XRS1		
QC-XRS2		

Table 3.4: Quality control standards and measurement conditions

NOTICE

The CONOSTAN reference sample bottles for DC and QC are labelled as “S-21+K+Sb”. The “+” character is not a valid character in SPECTRA.ELEMENTS names. Therefore, wherever this sample is referenced in the software, it is called “S21&K&Sb”.

Each of the QC samples has certified consensus values for PETRO-QUANT 4.1 BASIC analytes that the measured values are compared to. To measure the QC samples:

1. Open the “Control Specimens” tab of the LOADER. Upon choosing the PETRO-QUANT solution, all QC standards are automatically displayed and can then be moved to the correct position of the sample in the LOADER grid (see image below). The “Oil” application should be used. If you need a more detailed instruction for using the LOADER, refer to the SPECTRA.ELEMENTS software manual.

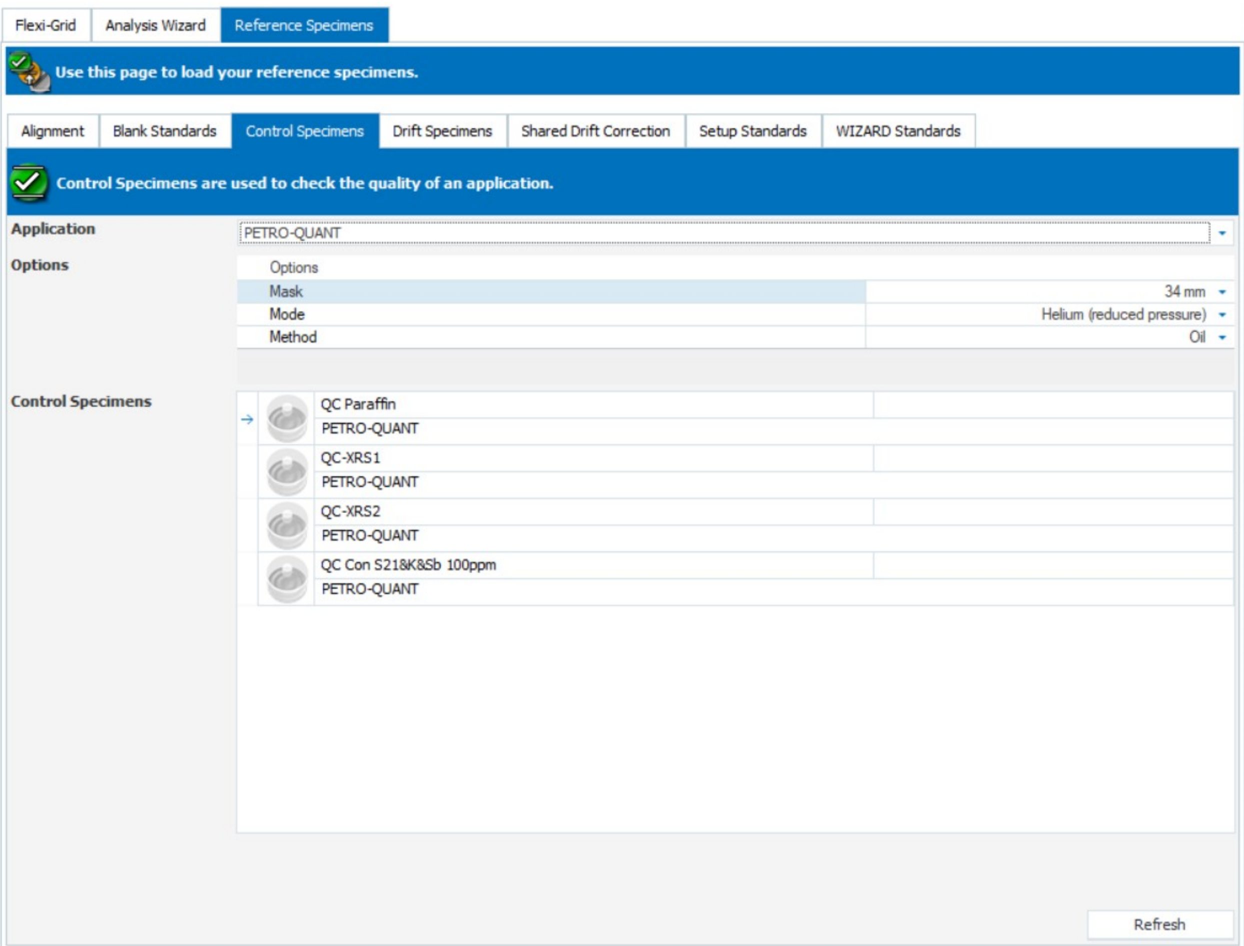

Figure 3.5: Image 6.1: QC Samples in the SPECTRA.ELEMENTS LOADER

2. Prepare each control sample individually according to the table above. Place cup 30s on a non-contaminating underlay for leakage testing, put liquid cup into sample holder with 34 mm aperture, place it on the respective Loader position and start measurement.
3. The SPECTRA.ELEMENTS software provides a quick check of the concentration values against accepted tolerances. The screen display automatically indicates the specification compliance by color. A green marker indicates that the result is within the tolerances defined in the Acceptance Protocol. Any mismatch is pointed out by red color (see below for what to do in this case).

If you are carrying out the first-time validation after installation, it is necessary to compare the measured concentration values after evaluation against the consensus values listed in the Acceptance Certificates.

3.2.3 What to Do if the QC Measurements Are Not Green

A failure of the validation procedure is indicated by a red marker for one or more elements in at least one of the standards. If the quality control fails, problems with the control samples, changes to the Flow Counter alignment, or an out-of-date drift correction could be the problem.

NOTICE

An S8-Check measurement should always be performed before validating PETRO-QUANT in order to ensure that the instrument is aligned properly. Refer to chapter *Performing an S8-Check Prior to Validating PETRO-QUANT [▶ 17]* and make sure that S8-Check is green before the measurement of the QC samples.

The following chart should help you troubleshoot a failed quality control sample, with each individual step described in more detail below:

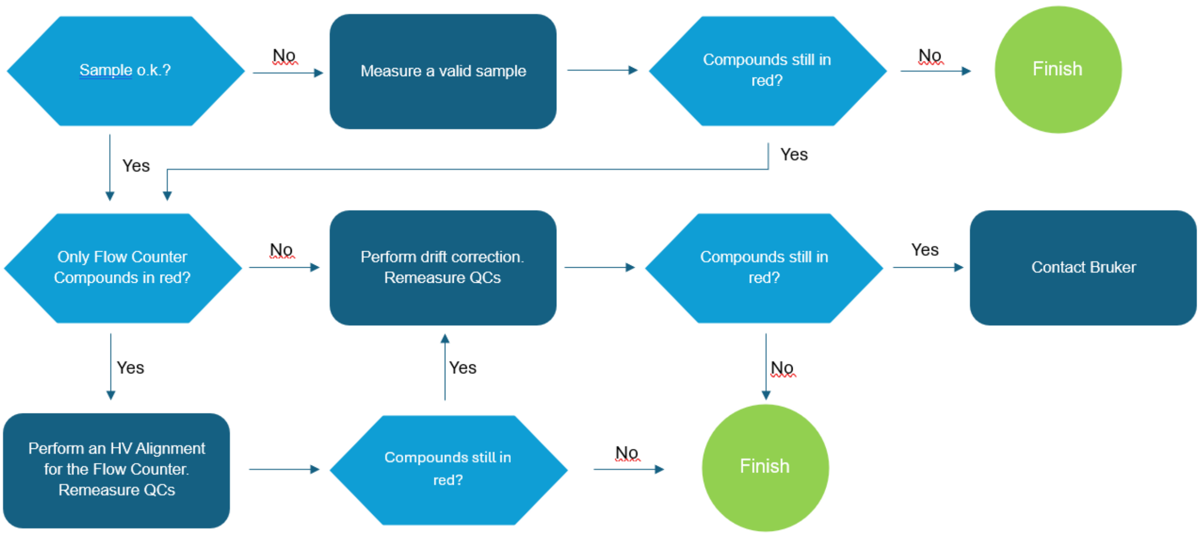

Figure 3.6: Failed QC Sample Decision Matrix

Checking the sample

The acceptance samples are all delivered in the same kind of HDPE or glass bottles. If you find a discrepancy running your quality check (QC) samples, please, check that QC sample and selected application match.

NOTICE

Never pour back samples to the original bottles! Some of the validation samples can be measured for setup purposes again, but we strongly recommend storing used material separately from the original samples. Samples with certified concentrations should never be measured twice if the exact concentrations are required.

Liquid samples have an expiration date. Typical values of stabilized commercial standard samples are 1-3 years unopened and 1-2 years opened. The lifetime of non-stabilized samples may be reduced to weeks and for very low concentrations these samples have to be used on the day of preparation.

If deviations occur running the QC samples, please, check the age of the samples. Order information for individual samples can be found in *Appendix 4: Order Information [▶ 44]*.

Furthermore, it is worth to perform a visual check for precipitation or the formation of layers. A homogenous sample is required for the analysis with PETRO-QUANT 4.1 BASIC.

Performing an HV alignment of the Flow Counter

Sometimes it is necessary to perform a HV alignment of the Flow Counter. The deviation will mainly be observed in the elements that are measured with the flow counter (in the case of PETRO-QUANT: Na, Mg, Al, Si, P, S, Cl, K, Ca, Ti, V, Cr, Mn, In, Sn, Sb, Ba). Typically the HV alignment of the flow counter should be performed after the change of Methane/Argon bottle.

What you need:

- The sample BAXS-ECO.

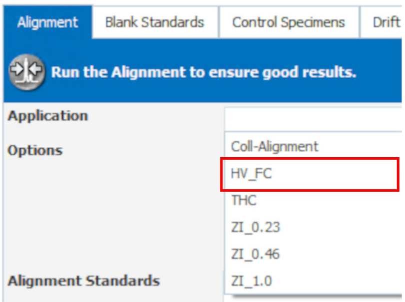

Figure 3.7: How to start the HV_FC alignment

1. Place the BAXS-ECO sample in the sample magazine.
2. In **LOADER**, select **Reference Specimens** > **Alignment**.
3. Select the application **HV_FC**.
4. Drag and drop the sample to the virtual loader.
5. Start the measurement.

Drift Correction (DC)

The intensities collected from the DC samples are used to routinely optimize your PETRO-QUANT 4.1 results.

During the initial validation of PETRO-QUANT 4.1 a complete set of drift correction measurements is run. If there was a problem with the analysis of these DC samples, this will cause errors with your QC data immediately afterwards.

If in the course of time the QC samples run out of specification limits this may indicate that it is time to repeat the drift correction.

If the quality control samples are outside of the specifications, the samples themselves are visually OK and not intermixed (see last two sections), and it is not an HV alignment issue of the flow counter, a drift correction is the typical measure. Refer to chapter *How to Perform a Drift Correction (Sensitivity) [▸ 12]*.

If only the blank sample QC measurement (QC Paraffin) fails, it can be sufficient to run the blank correction. If the blank QC sample is OK but at least one of the other QC samples fails, it may be sufficient to run the DC sample (DC Con S21&K&Sb 500ppm) only. However, we recommend running a complete drift correction set each time (exception: Running the Blank correction only is recommended after changing the foil batch for sample preparation if no other changes occurred).

After finishing the above steps, you can re-run your QC samples and compare the match of the values with the specifications indicated by the color markers of the results display.

If quality control still fails after all the steps suggested, contact your Bruker Service engineer.

3.2.4 Validating PETRO-QUANT Be

If you have the Be filter option in your instrument, your database comes with the additional solution version PETRO-QUANT Be, which has to be validated separately. The initial validation of PETRO-QUANT Be should always be carried out upon installation of the instrument, afterwards the validation can be performed according to need. PETRO-QUANT and PETRO-QUANT Be are validated separately, but according to the same procedure.

In order to validate PETRO-QUANT Be, follow the steps described in *How to Validate PETRO-QUANT 4.1 [▸ 18]*, but choose “PETRO-QUANT Be” as your solution in the LOADER Control Samples dialogue. The sample preparation and method choice is identical to the regular PETRO-QUANT validation procedure.

Note that due to the attenuating effect of the Be filter on the lower energy exciting radiation, the tolerance limits applied to Na and Mg are wider for PETRO-QUANT Be.

See also

🗎 What to Do if the QC Measurements Are Not Green [▸ 19]

4 PETRO-QUANT 4.1 Sample Preparation

PETRO-QUANT 4.1 BASIC uses just two different preparation procedures, one for oils (low volatility samples) and one for fuels (high volatility samples). One universal calibration is employed for all kinds of oils and fuels. Thus, the method requires a drift/mode correction system that allows running highly volatile samples with a calibration that has been set up with a different preparation/mode. The calibration reference is the application of a reduced pressure helium atmosphere and the preparation “PETRO-QUANT4-Oil” (see below). SPECTRA.ELEMENTS allows running samples prepared in different ways and measured in different modes by defining different applications in the solution and a few drift correction measurements. A second preparation for “PETRO-QUANT4-Fuel” has been set up this way (see below).

PETRO-QUANT4-Oil

The liquid cups are prepared using Prolene® film of 4 µm thickness. For each individual measurement, 7.0 ± 0.1 g of the oil sample are weighed in. The respective cup is placed 30 s on a non-contaminating underlay for leakage testing.

PETRO-QUANT4-Fuel

The liquid cups are prepared using Mylar® film of 3.6 µm thickness. For each individual measurement, 7.0 ± 0.1 g of the oil sample are weighed in. The respective cup is placed 30 s on a non-contaminating underlay for leakage testing.

NOTICE
Samples should always be prepared freshly before measurement. Leaving a prepared sample for too long before measurement can lead to foil sagging and influence measurement results. In the worst case, damage to the foil can occur, risking a leak of the oil sample into the measurement chamber.

5 PETRO-QUANT 4.1 Routine Analysis

PETRO-QUANT 4.1 BASIC can be used for direct analysis as well as for customer tailored methods and calibrations (see chapter *Customizing PETRO-QUANT 4.1 [▸ 29]*).

There are two basic applications for low-volatility ("PETRO-QUANT Oil") and high-volatility ("PETRO-QUANT Fuel") samples. Additionally, there are applications that can be used if the sample includes significant oxygen content, "PETRO-QUANT Oil Sulfonatic" and "PETRO-QUANT Aut-O-matic". The differences between the different applications and when to choose which will be described below (see chapters *Analysis of Samples with Low Volatility [▸ 25]*, *Analysis of Samples with High Volatility [▸ 25]* and *Analysis of Samples with Oxygen [▸ 26]*). All of these standard applications come with all preparation and measurement settings (foil, atmosphere mode, etc) stored.

5.1 How to Start an Unknown Sample Measurement

1. In the SPECTRA.ELEMENTS LOADER, select the FlexiGrid, choose "PETRO-QUANT" and select your atmosphere and application (see following sections for detailed description of the applications). All "Oil" based methods under reduced Helium ("Oil", "Oil Sulfonatic" and "Aut-O-matic") are based on the same measurement method, and thus, an erroneous selection of application at this stage can be resolved later. The sample can be re-evaluated assigning the correct application without running a new measurement.
2. Define a sample ID and edit the sample preparation settings according to the real values (see below). If you would like to measure duplicates of the same sample that are averaged on evaluation, you can right click the sample and click the "duplicate" button to create duplicates (or higher multiples by entering a different number than 2).
3. Prepare each sample individually. Place cup 30s on a non-contaminating underlay for leakage testing, put liquid cup into sample holder with 34 mm aperture, place it on the respective Loader position and start measurement.

NOTICE

Handling of duplicate measurements

If you measure duplicates in SPECTRA.ELEMENTS, the samples are usually placed into the LOADER and selected together. However, it is advisable to measure only one freshly prepared sample at a time. Before starting the measurement, make sure that only the one duplicate you would like to measure is selected, since clicking on duplicates usually selects all of them simultaneously.

Sample mass and density

For the correct evaluation of the sample, an accurate mass and density have to be provided to the software. The applications come with the following pre-defined preparation settings that have to be edited according to the real values:

	"Oil" applications (Oil, Oil Sulfonatic, Aut-O-matic)	"Fuel"
Density	0.87 g/cm^3	0.74 g/cm^3

	"Oil" applications (Oil, Oil Sulfonatic, Aut-O-matic)	"Fuel"
Sample mass	7.00 g	7.00 g

Table 5.1: Default sample mass and density for different applications

It is recommended to use a sample mass of 7.00 g, if this is not possible (e.g. due to lack of sufficient sample), the exact mass should be given for each sample. The other parameters such as foil and diameter of the sample cups are fixed in the respective application.

If the unknown sample has a considerably different density, PETRO-QUANT 4.1 BASIC allows a simple measurement and modification of the density. Thus, it includes a balance and disposable cups (volume 3.88 cm³ when filled to a flat surface at the top of the cup, i.e. no meniscus) to determine the respective density according to the following formula:

`ρ [g/cm³] = m [g] / V [cm³]`

here: `ρ [g/cm³] = m [g] / 3.81 cm³`

See *Appendix 3: Oil Density Determination [▸ 43]* for a table of sample mass values and the respective densities.

NOTICE
Measuring with the SampleCare option PETRO-QUANT Be
If your instrument includes the Be filter option, routine measurement using the PETRO-QUANT Be method follow the exact same procedure as for PETRO-QUANT. In order to choose the Be filter option, select "PETRO-QUANT Be" in the LOADERs FlexiGrid. The Be version can be useful to protect the tube in case of aggressive samples that are likely to cause foil damage. However, the analytical accuracy for lighter elements, especially Na and Mg, will be lower.

5.2 Analysis of Samples with Low Volatility

Samples with low volatility are typically analyzed by using the Helium (reduced pressure) mode and a Prolene® foil. The standard "PETRO-QUANT Oil" application defines CH2 as matrix compound. For samples with considerable oxygen contents please see chapter *Analysis of Samples with Oxygen [▸ 26]*).

5.3 Analysis of Samples with High Volatility

Samples with high volatility are typically analyzed using the Helium (atmospheric pressure) mode and a Mylar® foil.

"PETRO-QUANT Fuel" defines CH2 as matrix compound, the atmospheric pressure Helium measurement mode and the usage of a 3.6 µm Mylar® foil. If the measurement method is altered and fuel samples measured, the mode should never be changed to reduced pressure, as this can result in significant evaporation of the sample!

If the "Fuel" method is used for measurement, it is important that a blank sample measured for the "DC Fuel" method under Helium (atmospheric) mode (see *How to Perform a Blank Correction [▸ 14]*) has been measured with a Mylar® foil. Since the foil introduces contamination of several elements, a blank correction with the same foil always needs to be done to achieve accurate results.

See also

- Measurement in He atmosphere [▸ 9]

5.4 Analysis of Samples with Oxygen

There are two additional applications based on the “Oil” application that can be chosen if the sample contains oxygen:

The application “PETRO-QUANT Oil Sulfonatic” takes into account little oxygen contents calculated by stoichiometry from the sulfonatic sulfur content. The Conostan S21 standards with their sulfonatic composition are one example for oil samples that require such evaluation.

Lubricants with high concentrations of esters or used oils with significant amounts of water or alcohols require compensation of the oxygen’s absorption. The application “PETRO-QUANT Aut-O-matic” takes into account a significant oxygen content of the analyzed sample. Since the standard “Oil” application sets CH_2 as a matrix compound, high oxygen concentrations lead to less accurate results if not considered. This usually results in smaller concentrations of analyzed elements since oxygen absorbs a larger amount of fluorescent radiation than CH_2.

An insufficient matrix description by CH2 alone is also indicated in the interactive evaluation by the Compton coefficient which in “PETRO-QUANT Oil” should be very close to 1.00. A value higher than 1.05 indicates that the matrix contains heavier elements like oxygen (or additional elements not included in the PETRO-QUANT 4.1 BASIC method). In such cases, the “Aut-O-matic” application should be used for evaluation.

In the “PETRO-QUANT Aut-O-matic” application, the oxygen content is determined via the Compton coefficient. This requires the consideration of the correct density, mass and diameter of the sample. As mentioned above, the mass and density can be edited for each sample to reflect the true values of the individual sample. The diameter is fixed to the 3.5 cm inner diameter of the Bruker AXS liquid cup (order no. 7KP1901-8FA).

NOTICE

As the calculated oxygen content is determined via the Compton coefficient, it is recommended to pay attention to every parameter affecting it, i.e. sample mass and sample density. The more accurately these parameters are defined, the better the quality of the analysis results.

It is recommended to keep the sample mass at 7.00 g when measuring samples with the “Aut-O-matic” application. If low sample masses are used (especially below 5 g), small deviations in mass or density will cause large errors in the oxygen concentration determined via the Compton peak.

5.5 How to Perform an Interactive Evaluation

All samples measured with PETRO-QUANT 4.1 BASIC will be automatically evaluated using the application that was chosen for measurement. If you would like to change settings for an evaluation or choose a different application (e.g. change to Aut-O-matic if you suspect an oxygen content is influencing the results), this can be done in the SPECTRA.ELEMENTS XRF EVALUATION plugin. To evaluate a sample:

1. Select the sample in the Results Manager and select “Interactive Evaluation”
2. In the XRF EVALUATION plugin, select the application on the top right (do not select Fuel if the sample was measured with an Oil method and vice versa, as this will lead to wrong results!)
3. You can make changes to individual lines, for example in line selection and overlaps, by clicking on the dropdown menu for the line in the table. Re-evaluate after making changes by clicking on the calculator symbol.

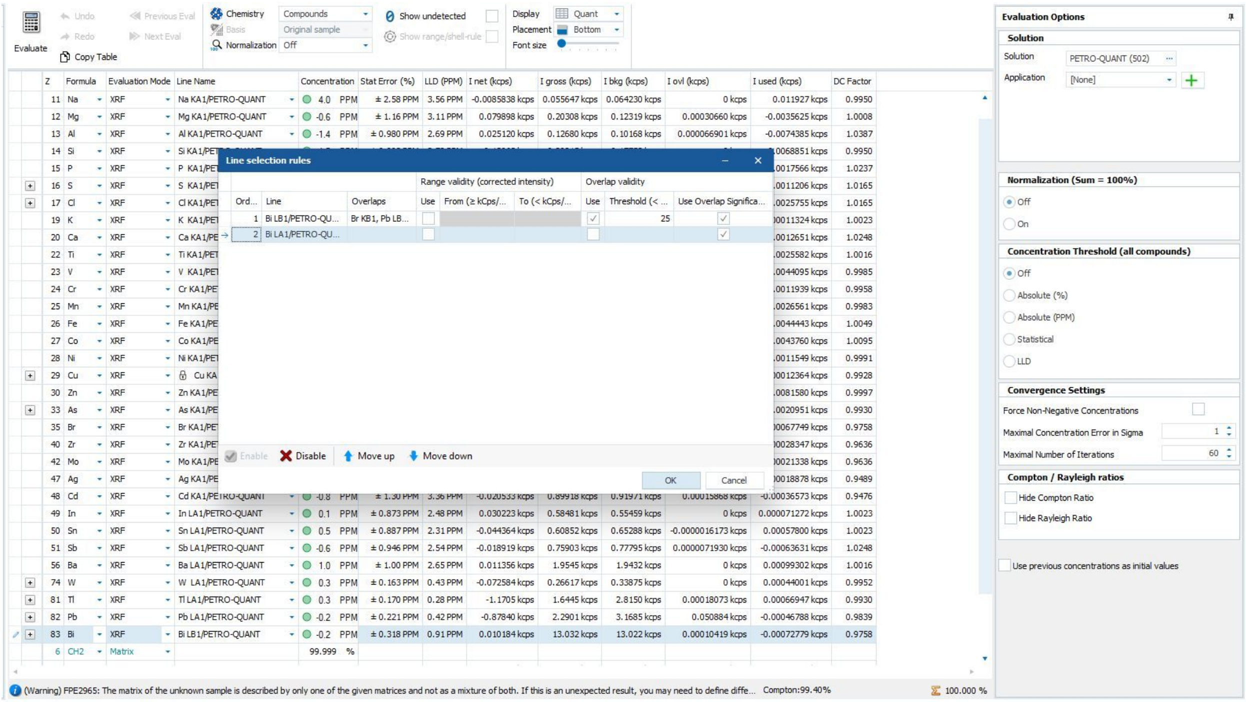

Figure 5.1: Editing line selection and overlap settings in interactive evaluation.

When evaluating automatically using the respective application, SPECTRA.ELEMENTS automatically chooses the line according to set selection criteria. There are different elements that have multiple lines in the PETRO-QUANT 4.1 BASIC measurement method:

- S and Cl: one line for trace analysis (KA1-Tr/PETRO-QUANT) and one line for minor analysis (KA1-Min/PETRO-QUANT). For Sulfur, an additional sulfonatic line (S KA1/ PETRO-QUANT-Sulfonatic) exists which is used for samples containing sulfonatic sulfur, as this induces a slight shift in peak position.
- As: two K lines (KA1/PETRO-QUANT and KB1/PETRO-QUANT)
- W, Tl, Pb, and Bi: two L lines (LA1/PETRO-QUANT and LB1/PETRO-QUANT)

The concentration based lines are selected according to a pre-determined concentration threshold. The trace lines are set as default in the respective evaluation model. When the trace region's intensity limit is exceeded, SPECTRA.ELEMENTS automatically activates the alternate –Min (minor) line for evaluation.

In case of different lines (KA1 and KB1 or respectively LA1 and LB1) set up to avoid severe line overlaps, SPECTRA.ELEMENTS allows to define a maximum overlapped intensity rate. Usually, the KA or LA lines are used by default due to higher intensity, and switched to KB or LB if the overlapped intensity exceeds a certain threshold.

However, in interactive evaluation, the lines can also be manually selected. PETRO-QUANT 4.1 BASIC also assists in creating specialist methods or calibrations, offering lines (pre-calibrated sets of measurement parameters) that are optimized for trace analysis in a light matrix and are already drift corrected (see *Customizing PETRO-QUANT 4.1 [▸ 29]*).

5.6 Troubleshooting Bad Evaluations

If your evaluation fails or you suspect that the results are faulty, there are some common scenarios where the issue can be resolved in evaluation without re-measuring the sample. For this, evaluate the sample using the interactive evaluation plugin (see section *How to Perform an Interactive Evaluation [▸ 26]*).

Poorly Defined Matrix

If you are measuring using the Oil application, but the Compton ratio is higher than 1 and you suspect that you have heavier elements than C in your matrix, you can switch to the Aut-O-matic application for evaluation. The matrix will then be modelled from the Compton peak by varying the oxygen concentration. This can be done directly in the evaluation plugin without measuring again.

Oscillations During Evaluation

If the evaluation did not converge due to oscillations, you will receive an oscillation warning already in the LOADER. This can often be resolved in the evaluation plugin without re-measuring.

The most common cause of oscillations are circular calculations (where multiple peaks overlap on each other and depend on each other). In this case, it is recommended to attempt switching a line in the high energy range (Pb, W, Bi, Tl) from the standard line to the other available line manually, as this often leads to a successful calculation. A common “problematic” overlap is Pb Lα1 and As Kα1, especially if both concentrations are low. Switching either of these lines to the corresponding β line resolves oscillations in most cases. It is also possible to set individual elements to a fixed concentration, for example to zero if you know your sample does not contain those elements. This decreases evaluation parameters and can lead to successful convergence.

6 Customizing PETRO-QUANT 4.1

The default PETRO-QUANT 4.1 BASIC is a universal analysis tool designed for a wide variety of petrochemical matrices and samples. It will handle all samples that fulfill the requirements presented in the second paragraph of chapter *Introduction [▸ 4]*.

However, for specific analytical tasks or samples that fall outside of the specifications it may be necessary to extend the capabilities of PETRO-QUANT by designing customized solutions.

There are three main approaches, depending on the analytical requirements:

- Editing the existing PETRO-QUANT method to customize measurement methods (useful for achieving lower limits of detection through extended counting times, lower measurement times and deactivating specific lines) or to create customized applications (for custom sample types and preparations or line selections).
- Importing PETRO-QUANT lines that are already calibrated and drift corrected into custom methods to combine with new lines.
- Creating custom calibrations based on PETRO-QUANT lines (but with new calibration and standards).

The latter two approaches make use of the “Shared Drift” concept, where a Shared Drift solution is created which contains the drift data for all solutions that use the same lines with the same measurement conditions.

NOTICE
The modification of solutions in SPECTRA.ELEMENTS requires a thorough understanding of the software structure and functionality. If you feel uncomfortable with possible risks of such advanced features, Bruker AXS offers so called “Application Support” packages to assist you in the creation of customized solutions.

If you want to check an entirely unknown sample that you suspect requires modifications to be measured with PETRO-QUANT, the standardless SMART-QUANT WD solution (delivered with your S8 Tiger Series 3) can be used. This is optimized to quickly analyze a wide variety of unknown samples to provide an overview without exact quantification. In order to measure an unknown oil sample:

1. Start by checking the preparation technique. Prepare an oil sample in a liquid cup with your foil and perform a leak test (see section *Sample Preparation in Liquid Cups [▸ 9]*).
2. Once you know the appropriate preparation, set up the measurement in SMART-QUANT WD. In the LOADER Flexi-Grid, choose the “Organics” application for SMART-QUANT WD. Choose the “Liquid” preparation and set your sample mass, density, foil and foil thickness.
3. Prepare each sample in a liquid cup according to your chosen preparation technique. Place cup 30s on a non-contaminating underlay for leakage testing, put liquid cup into sample holder with 34 mm aperture, place it on the respective Loader position and start measurement.

This measurement should be enough to verify whether the standard PETRO-QUANT method can be used, or whether modifications are needed. Specifically check for:

Poorly Defined Matrix

The standard matrix in the "Oil" application is CH2, same as in PETRO-QUANT. In some cases, this matrix description is not sufficient, which is indicated by a Compton ratio that differs from 1 (see also the chapter *Analysis of Samples with Oxygen [▸ 26]*). If the Compton ratio is above 1.05, this indicates that the matrix contains heavier elements like oxygen (or additional elements not included in the PETRO-QUANT 4.1 BASIC method).

How to approach: If you suspect the presence of oxygen, use the "Aut-O-matic" method to measure and evaluate samples. However, if you frequently measure the same sample type with the same oxygen content (within 0.5%), it is recommended to instead create a custom application with fixed oxygen content, since Aut-O-matic introduces more parameters and thus higher uncertainty into the evaluation. For this, move to section *How to Create Custom Applications [▸ 32]*.

Additional Elements

Is any element that is not covered by PETRO-QUANT 4.1 BASIC present?

How to approach: In order to analyze additional elements that are not included in PETRO-QUANT, you can create a new solution that combines PETRO-QUANT lines with one or more custom lines. To import lines from PETRO-QUANT you need to create a Shared Drift solution. See sections *Creating a Shared Drift Solution [▸ 34]* and *Importing Calibrated PETRO-QUANT Lines [▸ 38]*.

Additional Concentration Ranges

Is any element included in PETRO-QUANT 4.1 BASIC present in a concentration much higher than described in chapter *Introduction [▸ 4]* (so that the measurement parameters created for a trace (all) or minor (S and Cl) analysis are not applicable)?

How to approach: A new solution has to be created including your own calibration for the extended concentration range. Most of the time, the line parameters used in PETRO-QUANT can be adopted. This is done by creating a Shared Drift solution, which then automatically infers the line settings from PETRO-QUANT. See sections *Creating a Shared Drift Solution [▸ 34]* and *Reusing PETRO-QUANT Lines for Custom Calibrations [▸ 39]* for a description on how to create your own calibration based on a copy of a PETRO-QUANT line.

6.1 Customizing Measurement Methods and Applications

6.1.1 How to Create Custom Measurement Methods

In the measurement method, all variable parameters (i.e. parameters that do not affect the calibration) for the measurement are defined. This includes the mask and atmosphere mode, counting times for peak and backgrounds (or scan time in case of the Rh peak) and the measurement order. Lines can also be activated or deactivated for measurement. In PETRO-QUANT, two default measurement methods, "Oil" and "Fuel", are defined, which only differ in the He atmosphere used.

The counting times used in PETRO-QUANT 4.1 BASIC have been selected to produce best accuracy for the analysis of 32 elements within 20 minutes of total analysis time in various petrochemical matrices. For some samples it may be desirable to achieve shorter analysis times or lower detection limits. The latter can be done by increasing the counting times for peak and background positions of the respective lines in the PETRO-QUANT 4.1 measurement method (thus reducing statistical error and therefore improving LOD). Counting times can be adjusted for the respective applications in the "Measurement Methods" section of the SPECTRA.ELEMENTS WIZARD. Please refer to the SPECTRA.ELEMENTS user manual for a detailed description of how to do this.

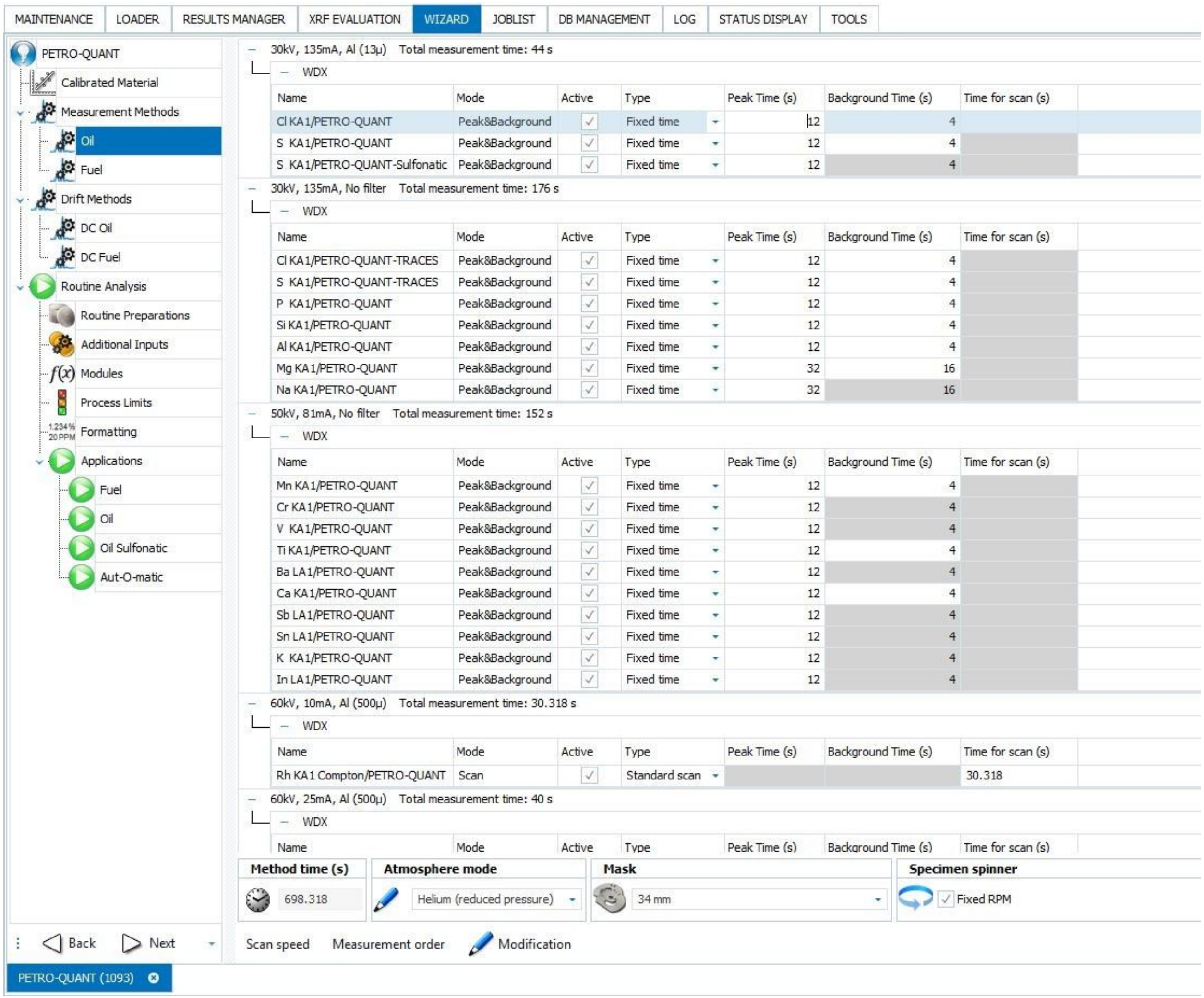

Figure 6.1: Changing the peak and background times of a line in the measurement method in the SPECTRA.ELEMENTS WIZARD

> **NOTICE**
>
> **Attention when changing measurement times**
>
> The film materials for preparation of the liquid cups have a limited lifetime when being irradiated by X-rays. Therefore, we recommend not to exceed a total measurement time of 30 minutes.

The requirement to prolong the measurement time for some element lines but not to exceed a 30 minutes measurement time might seem unsolvable at times. In this case the list of elements and lines can be reduced. For example, for specific analytical tasks the concentration of S and Cl is roughly known so that either the trace or the major line can be omitted. Furthermore, there might be some elements in the PETRO-QUANT 4.1 method that can be removed from the measurement method, since their measurement is not needed. Lines can be deactivated for future measurements in the “Measurement Methods” section of the SPECTRA.ELEMENTS WIZARD (see image above).

NOTICE

Attention when deactivating lines

All elements having an impact on matrix correction (i.e. elements with concentrations of more than several hundred µg/g - e.g. Sulfur) and all elements needed for overlap correction have to be measured to achieve a good result. If you are in doubt about overlaps, overlaps for each line can be checked in the “XRF Evaluation” plugin. In addition, all overlaps based on the line Rh KA1 Compton are calculated from its measured intensity. Therefore, do not skip the Compton line! Nevertheless, the elements only used for correction purposes can be analyzed with very short measuring times.

You can also create new measurement methods as a copy of an existing one by right clicking on its name in the WIZARD tree and selecting “Add new Measurement Methods”. It is recommended to preserve settings such as the measurement order, which is designed with potential decomposition and sedimentation risks in mind.

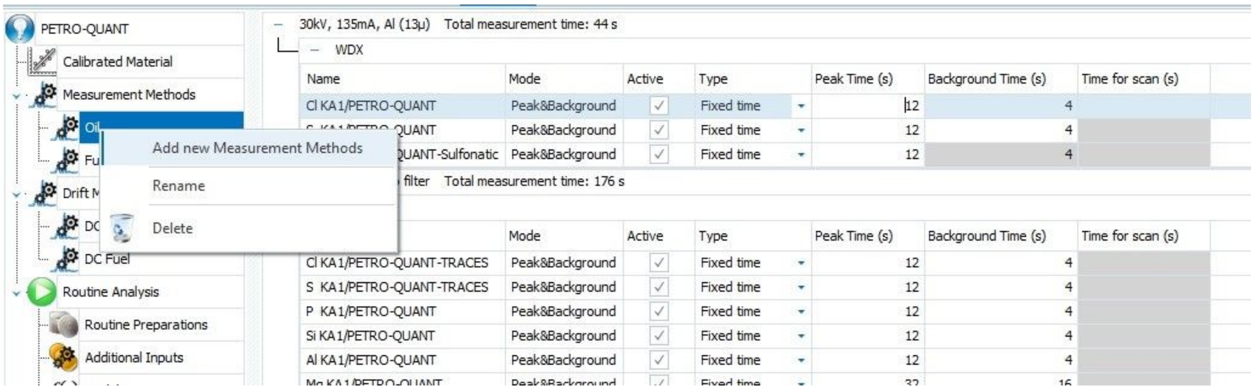

Figure 6.2: Copying a measurement method in WIZARD

If you create a new method, you need to ensure that an application exists that uses the new method. For this, see section *How to Create Custom Applications [▶ 32]*.

NOTICE

Drift and blank correction

In order to successfully measure unknowns, an appropriate drift and blank measurement must exist for your method. Modified measurement methods where counting times are changed or lines deactivated automatically use the drift entries measured with the standard method (“DC Oil” or “DC Fuel”). However, if you make changes to the atmosphere or mask (using another mask than 34 mm), you need to create a corresponding drift method under “Drift Methods” (“DC [MyMethodName]”) to measure the drift correction under the same conditions.

6.1.2 How to Create Custom Applications

Applications are used in SPECTRA.ELEMENTS to be able to define different types of samples or evaluation methods (e.g. with and without oxygen in the case of PETRO-QUANT). In each application, a combination of preparation and measurement method is chosen. Additionally, line selection rules can be chosen that only apply for this specific application. Applications are also used to set how the sample is evaluated, e.g. whether an oxygen concentration is calculated.

Applications can be customized in the respective node in the “Applications” section of the SPECTRA.ELEMENTS WIZARD. There are a few situations that make it necessary to modify PETRO-QUANT 4.1 BASIC applications:

If you have created a new measurement method (instead of modifying the standard “Oil” or “Fuel” methods), an application has to be set up to use this measurement method in order to make it available for measurements. In the “Methods” tab of the respective application section of the SPECTRA.ELEMENTS WIZARD, the measurement method for this application can be selected. You can also create a copy of an existing application by right clicking on its name in the WIZARD tree.

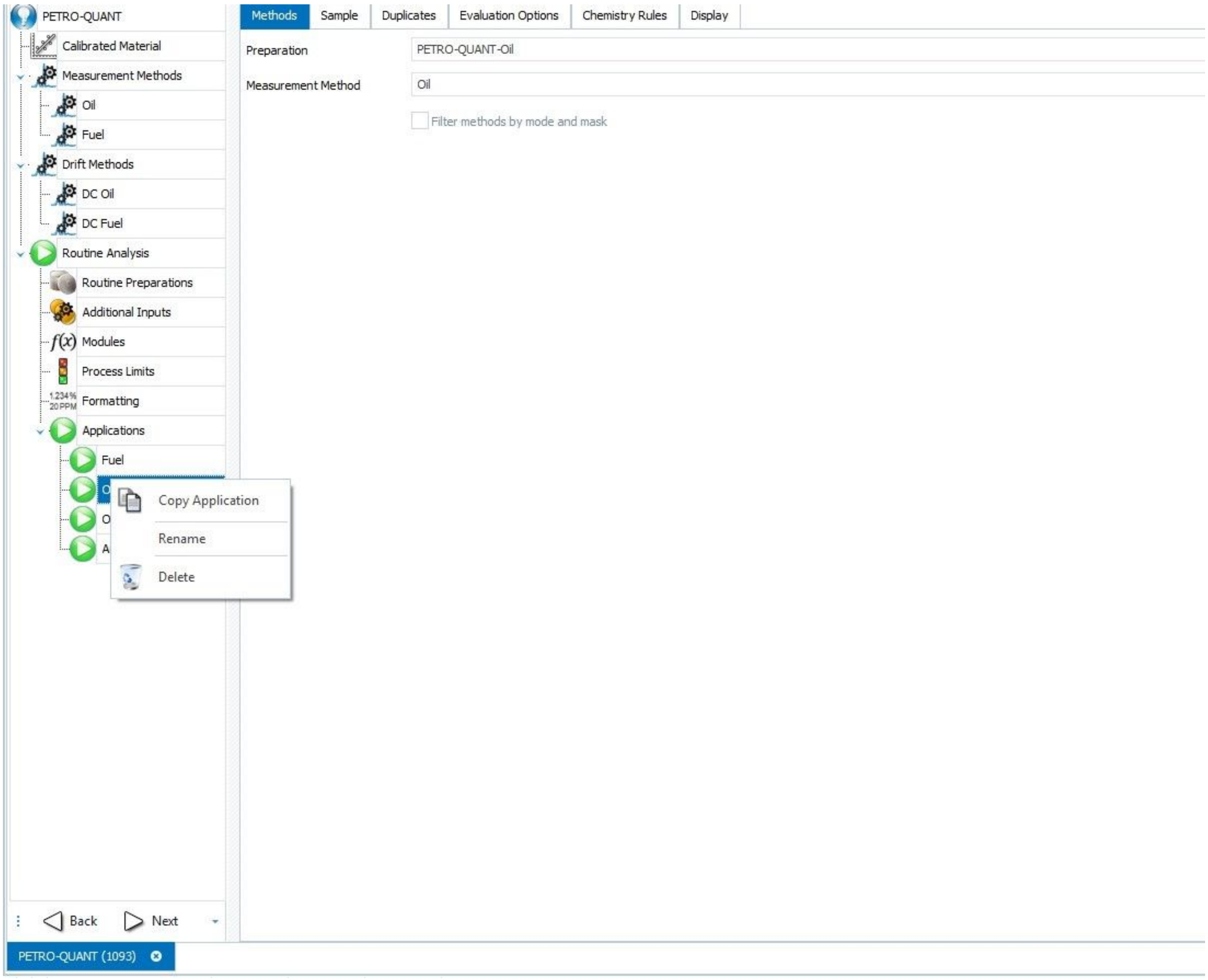

Figure 6.3: Copying an application in the WIZARD

For some real samples the default combination of low volatility sample and Prolene® film preparation may turn out to be insufficient. In case of unknown sample types it is worth testing the resistance of the film material before the analysis (see note at the end of chapter *Sample Preparation in Liquid Cups [▸ 9]*).

For a new preparation, create a new application by modifying a copy of the standard applications as described above. In the new application, the appropriate combination of preparation (depending on the film required), measurement method (Helium mode and analytical lines to be measured) and matrix compound can be selected. Additionally, changes to evaluation settings can be made here.

If you want to create an application for a sample type containing a typical amount of oxygen that does not vary much from sample to sample, it is recommended to set a fixed oxygen concentration rather than allowing calculation from the Compton ratio, as this reduces the evaluation parameters. This can be done in the “Chemistry Rules” tab. You can choose to set any element to a fixed concentration instead of evaluation by the XRF measurement.

> **NOTICE**
>
> **Drift and blank correction**
>
> In order to successfully measure unknowns, an appropriate drift and blank measurement must exist for your application. Blank measurements are usually used to offset the influence of preparation parameters such as the foil. However, the preparation is not defined in the BLANK sample since intensities are subtracted before any preparation correction takes place. If you create a custom preparation, always ensure that the latest BLANK measurement measured with your combination of measurement parameters (atmosphere + mask) matches your preparation.

If you create an application to measure matrices that significantly differ from the typical paraffin (e.g. fluff or polymers), the evaluation settings should be modified to enforce non-negative concentrations in order to produce optimal results with matrices different from the calibration matrix. This can be enabled under the “Evaluation” tab in the WIZARD section for the respective application. The “force non-negative concentrations” field should be checked for any application intended for use with a different matrix.

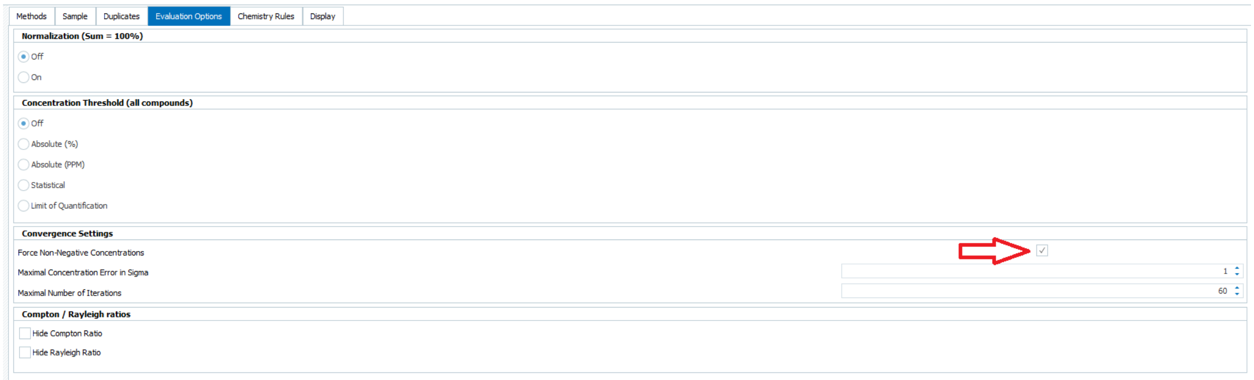

Figure 6.4: Select “Force non negative concentrations” in application with significantly different matrices.

After finishing setting up the applications, the new method has to be published (see above). Once the new method is published, its respective applications can be selected in the LOADER for measurement. The method is then locked for modification in WIZARD.

6.2 Custom Methods Based on PETRO-QUANT with Shared Drift

6.2.1 Creating a Shared Drift Solution

The analytical lines used in PETRO-QUANT 4.1 can be imported into custom solutions with their calibration intact or reused with a custom calibration (but keeping the line settings and drift correction of the original PETRO-QUANT). In SPECTRA.ELEMENTS, this requires the creation of a Shared Drift solution that will be used to automatically drift correct lines that are shared between methods. The following section describes how to create a Shared Drift solution. This procedure will also switch the default PETRO-QUANT solution to Shared Drift.

How to Create a Shared Drift Solution from PETRO-QUANT

When creating a Shared Drift solution from an existing method, this solution is then used in the future for drift correction, and the drift correction as well as the calibration of the lines contained in the Shared solution are automatically applied to all solution that use the lines. In order to create a Shared Drift solution:

1. Open the SPECTRA.ELEMENTS WIZARD and click the “Create a new Solution” in the upper left.

2. Choose “Shared Drift” as the solution type, and name your new solution “PETRO-QUANT Shared”.
3. On the main solution tab, click “Import Solution for Shared Drift”

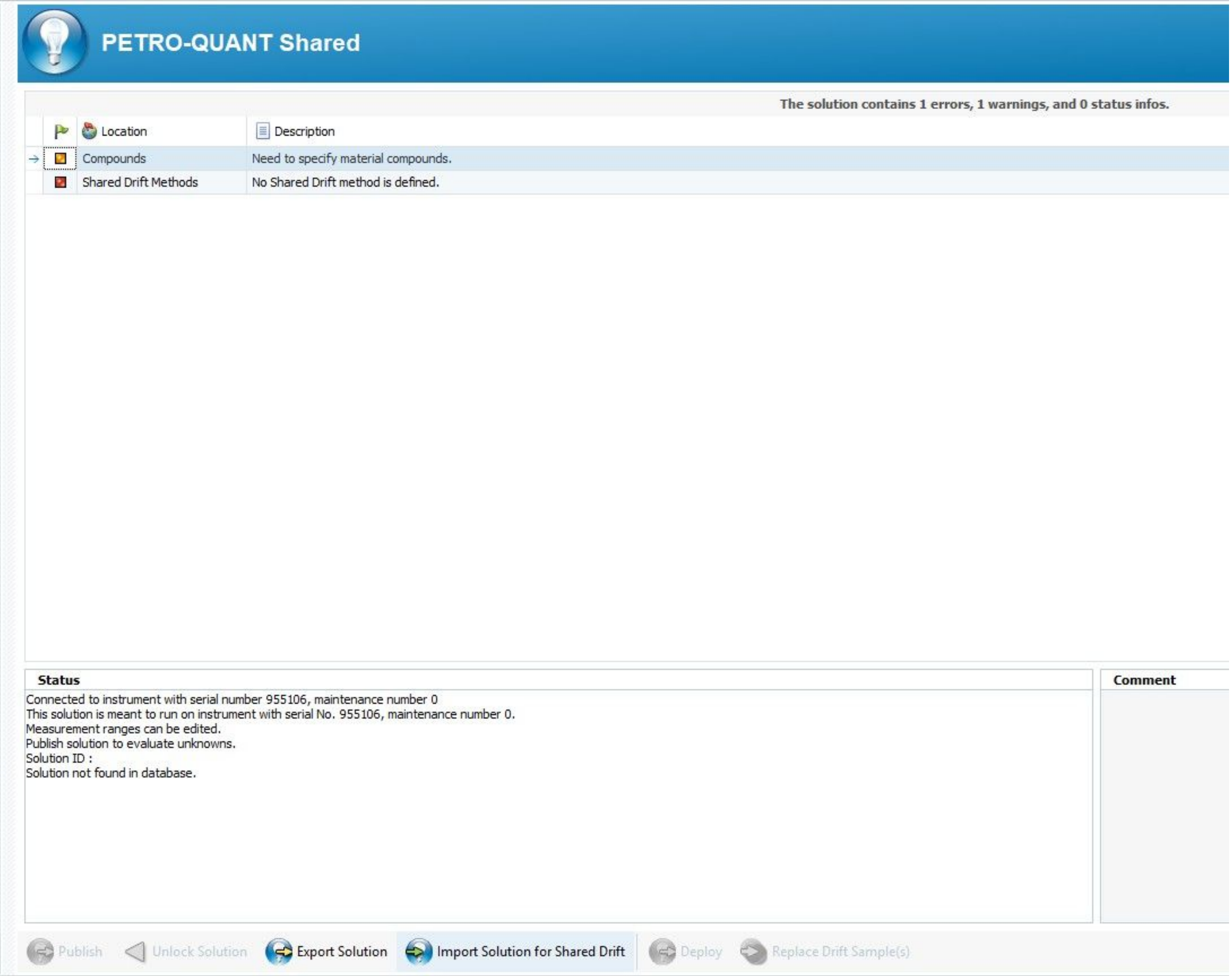

Figure 6.5: Creating a Shared Drift solution based on PETRO-QUANT.

4. Select PETRO-QUANT. The drift standards used in in PETRO-QUANT appear.
5. Click on Import.
6. Publish the solution.

The lines contained in the Shared solution can now be used in other methods. However, this means that drift correction is now handled via the Shared solution.

NOTICE

Shared drift correction

When you have created a Shared solution from PETRO-QUANT, you need to drift correct using the Shared solution. This ensures that the lines are drift corrected in every method that make use of the PETRO-QUANT lines (including the standard PETRO-QUANT).

In order to drift correct a Shared solution, choose the “Shared Drift Correction” tab in the LOADER instead of the “Drift Specimens” tab. Then choose the “PETRO-QUANT Shared” solution. After choosing the drift method (“DC Oil” or “DC Fuel”), the usual drift samples should appear, and the drift correction can be continued according to chapter Drift Correction *Drift Correcting PETRO-QUANT 4.1 [▸ 12]*.

6.2.2 Adding PETRO-QUANT to an Existing Shared Drift

If you would like to combine existing, calibrated analytical lines from PETRO-QUANT with another method such as SMART-QUANT, the easiest way is to add both methods to the same Shared Drift.

In order to do this, use the following steps:

1. Open the target Shared Drift solution, e.g. SMART-QUANT Shared, from the database in the SPECTRA.ELEMENTS WIZARD.
2. Unlock the solution. An option for importing a solution into the Shared Drift should now appear on the bottom:

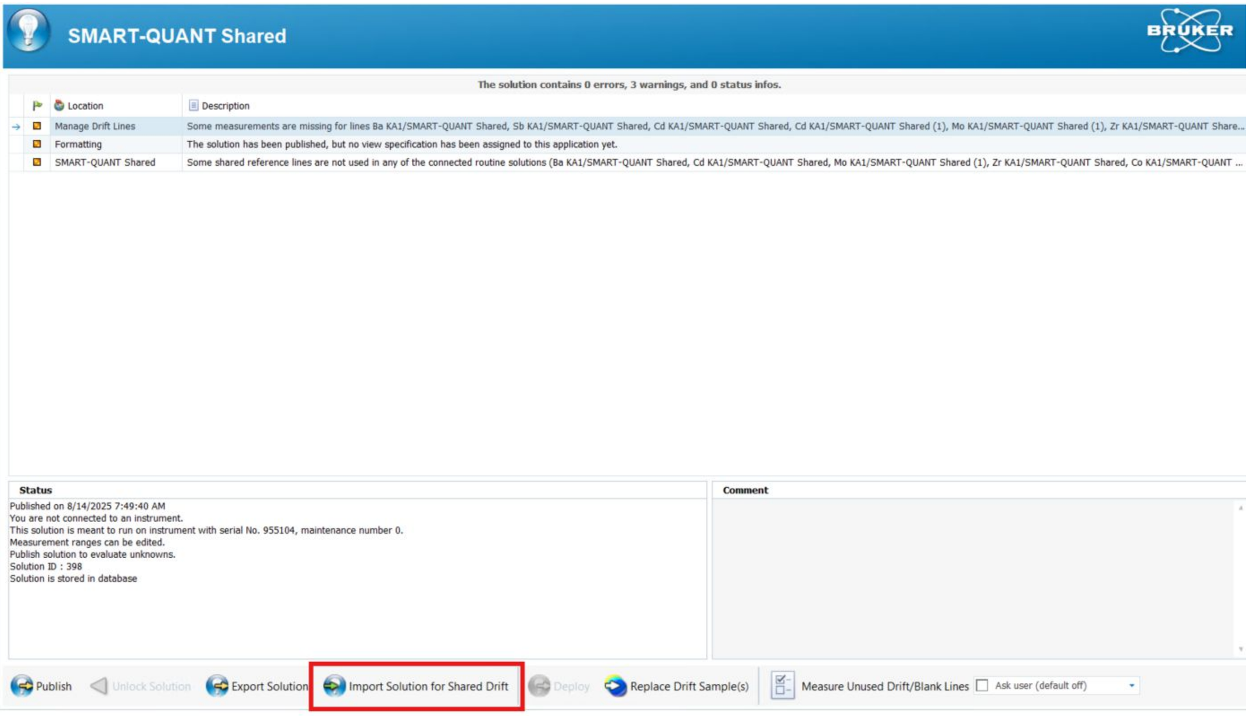

Figure 6.6: Import Solution to Shared Drift

3. After clicking, the solution list appears. Select the latest published version of the PETRO-QUANT method that you want to switch to Shared Drift from the list.
4. A dialogue with the drift standards is opened. The default option should be to import the PETRO-QUANT oil standard and use that one for the imported drift lines from PETRO-QUANT. Ensure that **only** this standard is selected, then click “Import”.

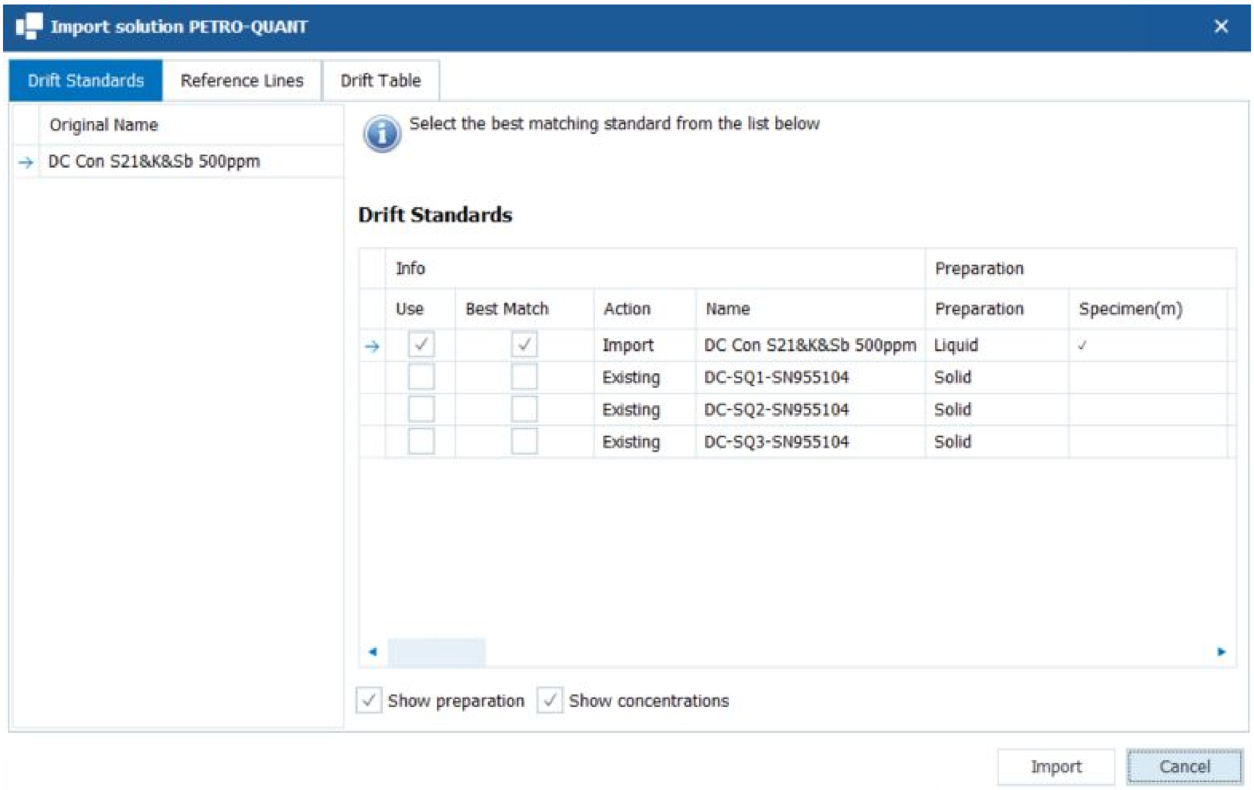

Figure 6.7: Import PETRO Drift Standard to Shared Drift

5. The new drift standard should now appear in “Shared Drift Standards”. The PETRO drift lines with the oil drift standards should appear in “Line Settings”.
6. Check the drift methods. The PETRO lines should be active for the measurement methods “Drift 34mm Helium Reduced” and “Drift 34mm Helium Atmospheric”. If not, activate the lines in these measurement methods.
7. Check the main Shared Drift solution node. The option “Measure Unused Lines” should be changed to “Force Off”. This will ensure that for the measurements of the PETRO drift sample, only the lines that use this sample as Shared Drift are measured.

NOTICE

Measure Unused Lines

If you create a Shared Drift method which comprises multiple methods with different drift samples (e.g. SMART QUANT + PETRO QUANT), contrary to the original PETRO-QUANT method, Measure Unused Lines should NEVER be turned on. If this option is turned off, only the drift lines belonging to the current drift sample are measured. If the DC Con S21&K&Sb 500ppm sample is chosen in the LOADER, only the PETRO lines will be measured. Accidentally selecting the Measure Unused Lines option can lead to very large measurement times that can damage liquid samples! Thus, it is recommended to set this option to “Force Off” in Shared Drift solutions.

In Shared Drift, PETRO lines that are not used in the drift correction are not imported. This means, that the measurement will be slightly faster than the original PETRO drift, and slight systematic deviations may occur in the drift factors for light elements. If you notice problems with validation after switching to Shared Drift, check the drift measurement method: The measurement order should resemble the original PETRO QUANT order (starting with S, P then the rest of the light elements, then Ag, Cd and Compton). The measurement time of the first lines can be increased to more closely match the original order if the drift factors do not match the observed difference in intensity in the QC sample.

6.2.3 Importing Calibrated PETRO-QUANT Lines

Once you have created a Shared solution from PETRO-QUANT, you can import lines into your own custom solutions which will be automatically drift corrected via PETRO-QUANT Shared. To import PETRO lines into your own custom solution:

1. Open the Compounds tab of your new solution
2. Click “Import Lines”
3. Choose PETRO-QUANT in the Solutions dropdown menu. On the right side you should see the label "Using shared drift PETRO-QUANT Shared”
4. Choose the lines that you want to import for your new solution
5. Click on “Import Lines”

NOTICE

Make sure that you include any compounds that have overlaps defined (either as measured lines, or as compounds with fixed concentrations) or turn off the respective overlaps in your application under the “Chemistry Rules” tab (see *How to Create Custom Applications [▸ 32]*). It is also recommended to always include the Compton line.

Then you can add your own additional elements. The respective lines can be customized in the “Analytical Lines” tab. For the imported lines, the calibration is locked and the drift table will be imported from the Shared solution.

The PETRO-QUANT lines (except the Compton line) use a Blank measurement in the calibration. Valid blank data is therefore required in order to evaluate measurements with the new method. A blank standard with matching concentrations will be created automatically when you import lines. Since the set of lines in your custom solution is different from the original PETRO-QUANT and may include additional compounds, you have to measure the Blank once before you can publish the new solution. After you have imported the lines, follow these steps for the Blank measurement:

1. Open the “Manage Standards” tab. The Blank sample will be visible here (with three duplicates as in the original PETRO-QUANT), along with any standards you have added for your own custom lines (PETRO-QUANT calibration standards are not accessible since the calibration is locked).
2. Click on “Measure in LOADER”. This will switch you to the LOADER plugin.
3. Place the Blank samples in the LOADER grid. Prepare each sample with 7.00g Paraffin and the Prolene® foil.
4. After measuring the Blanks, go to the “Calibrate Lines” node and click through the lines. Make sure that the lines imported from PETRO-QUANT (except the Compton line, which does not use a Blank) have “Use Blank” selected.
5. Publish your solution in the main solution node. Now it is possible to measure the Blank samples using the “Blank Specimen” tab in the LOADER. You can also define an application using atmospheric Helium, for which you have to measure a Blank sample using the “DC Fuel” method.

Drift measurements for your new solution are performed using Shared Drift. If you have added compounds that are not part of the PETRO-QUANT package and thus are not part of the Shared Drift, you can either manually assign a reference line from the Shared Drift (a neighboring line with the same excitation conditions) or add the line to Shared Drift by clicking the “Add Line to Shared Drift” button in Analytical Lines.

In order to be able to measure the Shared Drift drift specimen, you need to select the “Shared Drift Correction” tab in the LOADER. If you have created a customized solution, a Shared Drift correction will apply to both PETRO-QUANT and your custom solution.

NOTICE

Copying Lines to Shared Drift

Note that when you copy a line to Shared Drift, you need to ensure that the Shared Drift solution is modified to accommodate the new line. Most importantly, a drift standard has to be defined that contains the new element, and assigned to the new line.

6.2.4 Reusing PETRO-QUANT Lines for Custom Calibrations

Once you have created the Shared Drift solution PETRO-QUANT Shared, you can easily repurpose the PETRO-QUANT lines for your own custom calibrations, using the Shared Drift for drift correction. This is useful when you want to use the same line settings and drift correction, but also create your own calibration with different standards, for example to use a different preparation or longer measurement time during calibration to achieve better analytical results.

In order to reuse lines with custom calibration in your own method:

1. Create your own WDX analysis solution in the SPECTRA.ELEMENTS WIZARD plugin.
2. In the “Calibrated Materials” tab, on the bottom of the page, an option is available to select a Drift solution. In the drift solution, select Shared Drift and “PETRO-QUANT Shared”.

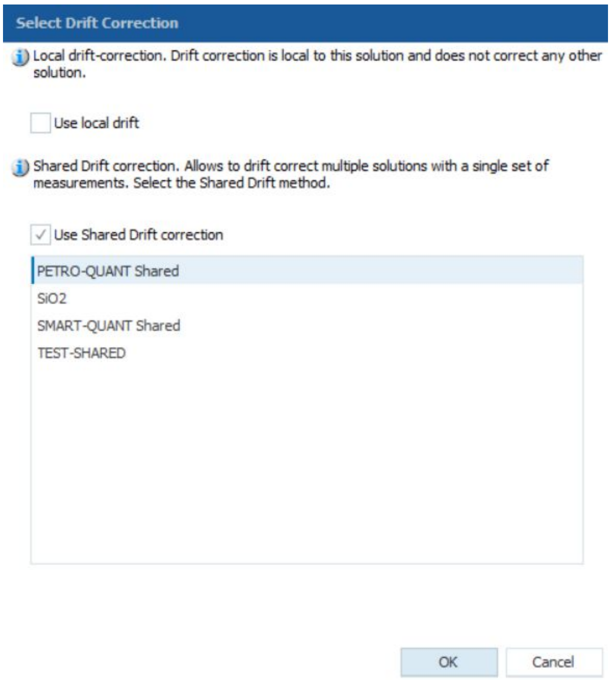

Figure 6.8: Setting the Drift Solution to “PETRO-QUANT Shared”

3. Now set up your method: Choose the compounds, create standards and preparations.
4. Go to “Analytical Lines”: For all compounds that are part of the PETRO-QUANT package, the lines that are created will automatically be compatible with the Shared Drift. This means that the excitation conditions and settings will match those from PETRO-QUANT, which are already optimized for trace analysis in a hydrocarbon matrix. All lines for compounds that are part of PETRO-QUANT will also have a reference line assigned for drift automatically from the Shared solution.

5. If you have added compounds that are not part of the PETRO-QUANT package and thus are not part of the Shared Drift, you can either manually assign a reference line from the Shared Drift (a neighboring line with the same excitation conditions) or add the line to Shared Drift by clicking the “Add Line to Shared Drift” button in Analytical Lines.

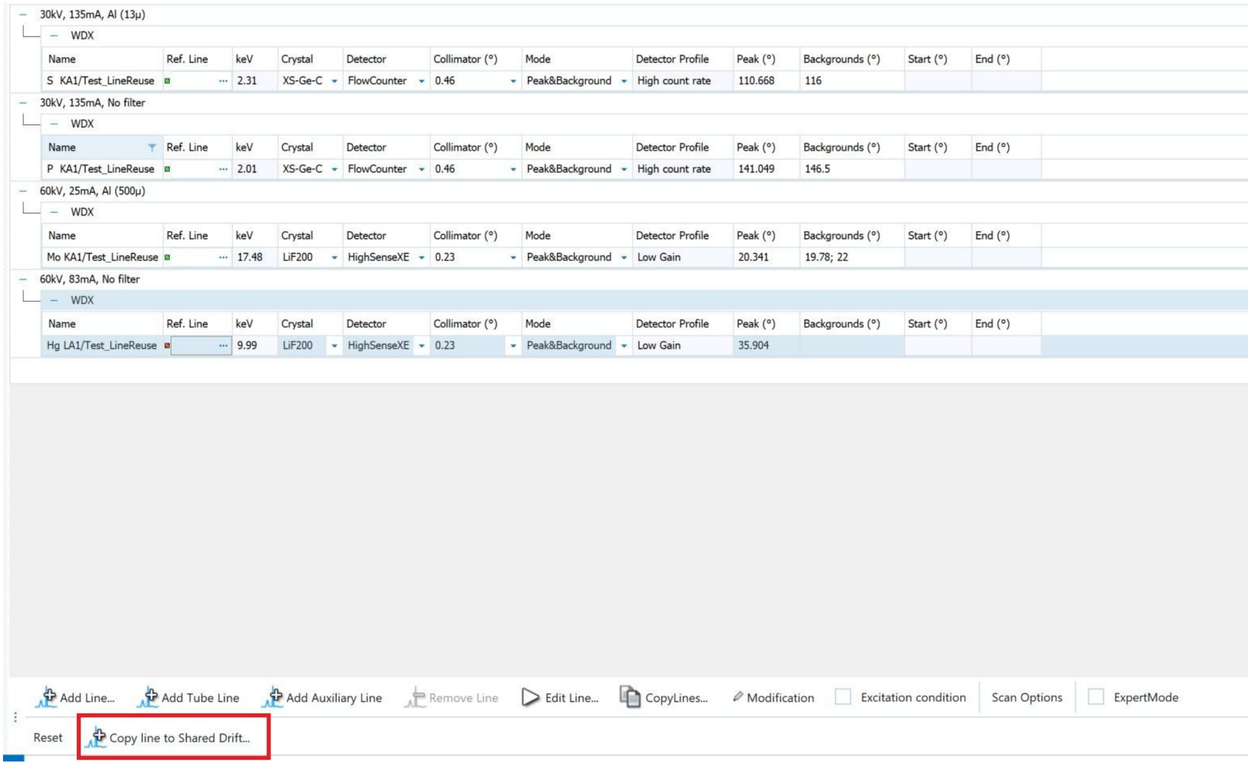

Figure 6.9: Copying a new line to Shared Drift

NOTICE

Copying Lines to Shared Drift

Note that when you copy a line to Shared Drift, you need to ensure that the Shared Drift solution is modified to accommodate the new line. Most importantly, a drift standard has to be defined that contains the new element, and assigned to the new line.

The calibration method (containing settings such as measurement times for each line) is fully customized in your new solution. You can measure your own calibration standards and create your own calibration for the reused lines. Refer to the SPECTRA.ELEMENTS user manual for more information on how to create a calibration in the WIZARD.

7 Appendix

7.1 Appendix 1: Back-Up and Re-Installation of the PETRO-QUANT Method

Your S8 Tiger Series 3 is delivered with PETRO-QUANT 4.1 factory installed onto the device database. All changes to the method and applications can be easily saved to the database, creating a new version of PETRO-QUANT.

The database system conveniently enables the preservation of all methods, method versions over time, and measurement data. The instrument database is preserved even in the event of a new installation of the SPECTRA.ELEMENTS software, as it is independent of the software and not overwritten upon software re-installation.

NOTICE

We strongly recommend that you make regular back-up copies of the device database. This will enable you to access previous versions of the method and measurement data in case of problems.

A database backup can be performed in the DB Management plugin in the "General Settings" tab. The SPECTRA.ELEMENTS software manual contains information regarding how to perform a database backup and how to handle databases in general.

If for some reason you need to reset the database and re-install PETRO-QUANT, this can be done by opening the .bsml file in the SPECTRA.ELEMENTS WIZARD, Unlocking the method, choosing "Adapt to current hardware" and then publishing the method. In the case of re-installation, the drift and blank samples of the "Oil" and "Fuel" method have to be measured at least once before the method can be used. Refer to the section *Drift Correcting PETRO-QUANT 4.1 [▸ 12]*.

7.2 Appendix 2: PETRO-QUANT 4.1 Reference Samples

	Na [mg/kg]	Mg [mg/kg]	Al [mg/kg]	Si [mg/kg]	P [mg/kg]	S [%]	Cl [mg/kg]
QC-Paraffin	0	0	0	0	0	0.0000	0
Alignment-CON S21&K&Sb 500 ppm	500	500	500	500	500	*[1.28]*	
Alignment-XRS1	900					0.0500	10000
Alignment-XRS2						*[1.24]*	
QC-CON S21&K&Sb 100 ppm	100	100	100	100	100	*[0.255]*	
QC-XRS1						0.0100	100
QC-XRS2						*[0.058]*	
[ ] informational values							
	K [mg/kg]	**Ca [mg/kg]**	**Ti [mg/kg]**	**V [mg/kg]**	**Cr [mg/kg]**	**Mn [mg/kg]**	**Fe [mg/kg]**
QC-Paraffin	0	0	0.0	0.0	0.0	0.0	0.0

	Na [mg/kg]	Mg [mg/kg]	Al [mg/kg]	Si [mg/kg]	P [mg/kg]	S [%]	Cl [mg/kg]
Alignment-CON S21&K&Sb 500 ppm	500	500	500.0	500.0	500.0	500.0	500.0
Alignment-XRS1							
Alignment-XRS2							
QC-CON S21&K&Sb 100 ppm	100	100	100.0	100.0	100.0	100.0	100.0
QC-XRS1							
QC-XRS2							
	Co [mg/kg]	**Ni [mg/kg]**	**Cu [mg/kg]**	**Zn [mg/kg]**	**As [mg/kg]**	**Br [mg/kg]**	**Zr [mg/kg]**
QC-Paraffin	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Alignment-CON S21&K&Sb 500 ppm		500.0	500.0	500.0			
Alignment-XRS1						350.0	350.0
Alignment-XRS2	250.0				350.0		
QC-CON S21&K&Sb 100 ppm		100.0	100.0	100.0			
QC-XRS1						100.0	100.0
QC-XRS2	100.0				100.0		
	Mo [mg/kg]	**Ag [mg/kg]**	**Cd [mg/kg]**	**In [mg/kg]**	**Sn [mg/kg]**	**Sb [mg/kg]**	**Ba [mg/kg]**
QC-Paraffin	0.0	0	0	0	0	0	0
Alignment-CON S21&K&Sb 500 ppm	500	500	500		500	500	500
Alignment-XRS1							
Alignment-XRS2				500			
QC-CON S21&K&Sb 100 ppm	100.0	100	100		100	100	100
QC-XRS1							
QC-XRS2				100			
	W [mg/kg]	**Tl [mg/kg]**	**Pb [mg/kg]**	**Bi [mg/kg]**	**CH2 [%]**	**O [%]**	**ρ [g/cm³]**
QC-Paraffin	0.0	0.0	0.0	0.0	100.0000	0.0	0.86
Alignment-CON S21&K&Sb 500 ppm			500.0			*[1.91]*	0.90
Alignment-XRS1						1.03	0.87
Alignment-XRS2	500.0	250.0		250.0		0.334	0.87
QC-CON S21&K&Sb 100 ppm			100.0			*[0.382]*	0.87
QC-XRS1						1.00	0.85

	Na [mg/kg]	Mg [mg/kg]	Al [mg/kg]	Si [mg/kg]	P [mg/kg]	S [%]	Cl [mg/kg]
QC-XRS2	100.0	100.0		100.0		*[0.088]*	0.90
[ ] informational values							

Table 7.1: Overview of reference sample concentrations

7.3 Appendix 3: Oil Density Determination

Sample Mass* [g]	Density [g/cm³]		Sample Mass* [g]	Density [g/cm³]		Sample Mass* [g]	Density [g/cm³]
2.51 | 2.52 | 2.53 | 2.54	0.65		2.55 | 2.56 | 2.57 | 2.58	0.66		2.59 | 2.60 | 2.61	0.67
2.62 | 2.63 | 2.64 | 2.65	0.68		2.66 | 2.67 | 2.68 | 2.69	0.69		2.70 | 2.71 | 2.72 | 2.73	0.70
2.74 | 2.75 | 2.76 | 2.77	0.71		2.78 | 2.79 | 2.80 | 2.81	0.72		2.82 | 2.83 | 2.84 | 2.85	0.73
2.86 | 2.87 | 2.88 | 2.89	0.74		2.90 | 2.91 | 2.92	0.75		2.93 | 2.94 | 2.95 | 2.96	0.76
2.97 | 2.98 | 2.99 | 3.00	0.77		3.01 | 3.02 | 3.03 | 3.04	0.78		3.05 | 3.06 | 3.07 | 3.08	0.79
3.09 | 3.10 | 3.11 | 3.12	0.80		3.13 | 3.14 | 3.15 | 3.16	0.81		3.17 | 3.18 | 3.19 | 3.20	0.82
3.21 | 3.22 | 3.23	0.83		3.24 | 3.25 | 3.26 | 3.27	0.84		3.28 | 3.29 | 3.30 | 3.31	0.85
3.32 | 3.33 | 3.34 | 3.35	0.86		3.36 | 3.37 | 3.38 | 3.39	0.87		3.40 | 3.41 | 3.42 | 3.43	0.88
3.44 | 3.45 | 3.46 | 3.47	0.89		3.48 | 3.49 | 3.50 | 3.51	0.90		3.52 | 3.53 | 3.54 | 3.55	0.91
3.56 | 3.57 | 3.58	0.92		3.59 | 3.60 | 3.61 | 3.62	0.93		3.63 | 3.64 | 3.65 | 3.66	0.94
3.67 | 3.68 | 3.69 | 3.70	0.95		3.71 | 3.72 | 3.73 | 3.74	0.96		3.75 | 3.76 | 3.77 | 3.78	0.97
3.79 | 3.80 | 3.81 | 3.82	0.98		3.83 | 3.84 | 3.85 | 3.86	0.99		3.87 | 3.88 | 3.89	1.00
3.90 | 3.91 | 3.92 | 3.93	1.01		3.94 | 3.95 | 3.96 | 3.97	1.02		3.98 | 3.99 | 4.00 | 4.01	1.03
4.02 | 4.03 | 4.04 | 4.05 |	1.04		4.06 | 4.07 | 4.08 | 4.09 |	1.05		4.10 | 4.11 | 4.12 | 4.13	1.06
4.14 | 4.15 | 4.16 | 4.17	1.07		4.18 | 4.19 | 4.20	1.08		4.21 | 4.22 | 4.23 | 4.24	1.09
4.25 | 4.26 | 4.27 | 4.28	1.10		4.29 | 4.30 | 4.31 | 4.32	1.11		4.33 | 4.34 | 4.35 | 4.36	1.12
4.37 | 4.38 | 4.39 | 4.40	1.13		4.41 | 4.42 | 4.43 | 4.44	1.14		4.45 | 4.46 | 4.47 | 4.48	1.15

Sample Mass* [g]	Density [g/cm^3]		Sample Mass* [g]	Density [g/cm^3]		Sample Mass* [g]	Density [g/cm^3]
4.49 | 4.50 | 4.51 | 4.52	1.16		4.53 | 4.54 | 4.55	1.17		4.56 | 4.57 | 4.58 | 4.59	1.18
4.60 | 4.61 | 4.62 | 4.63	1.19		4.64 | 4.65 | 4.66 | 4.67	1.20		4.68 | 4.69 | 4.70 | 4.71	1.21
4.72 | 4.73 | 4.74 | 4.75	1.22		4.76 | 4.77 | 4.78 | 4.79	1.23		4.80 | 4.81 | 4.82 | 4.83	1.24
4.84 | 4.85 | 4.86	1.25		4.87 | 4.88 | 4.89 | 4.90	1.26		4.91 | 4.92 | 4.93 | 4.94	1.27
4.95 | 4.96 | 4.97 | 4.98	1.28		4.99 | 5.00 | 5.01 | 5.02	1.29		5.03 | 5.04 | 5.05 | 5.06	1.30
5.07 | 5.08 | 5.09 | 5.10	1.31		5.11 | 5.12 | 5.13 | 5.14	1.32		5.15 | 5.16 | 5.17	1.33
5.18 | 5.19 | 5.20 | 5.21	1.34		5.22 | 5.23 | 5.24 | 5.25	1.35		5.26 | 5.27 | 5.28 | 5.29	1.36
5.30 | 5.31 | 5.32 | 5.33	1.37		5.34 | 5.35 | 5.36 | 5.37	1.38		5.38 | 5.39 | 5.40 | 5.41	1.39
5.42 | 5.43 | 5.44 | 5.45	1.40		5.46 | 5.47 | 5.48 | 5.49	1.41		5.50 | 5.51 | 5.52	1.42
5.53 | 5.54 | 5.55 | 5.56	1.43		5.57 | 5.58 | 5.59 | 5.60	1.44		5.61 | 5.62 | 5.63 | 5.64	1.45
5.65 | 5.66 | 5.67 | 5.68	1.46		5.69 | 5.70 | 5.71 | 5.72	1.47		5.73 | 5.74 | 5.75 | 5.76	1.48
5.77 | 5.78 | 5.79 | 5.80	1.49		5.81 | 5.82 | 5.83	1.50		5.84 | 5.85 | 5.86 | 5.87	1.51
5.88 | 5.89 | 5.90 | 5.91	1.52		5.92 | 5.93 | 5.94 | 5.95	1.53		5.96 | 5.97 | 5.98 | 5.99	1.54
6.00 | 6.01 | 6.02 | 6.03	1.55		6.04 | 6.05 | 6.06 | 6.07	1.56		6.08 | 6.09 | 6.10 | 6.11	1.57
6.12 | 6.13 | 6.14	1.58		6.15 | 6.16 | 6.17 | 6.18	1.59			
* Sample mass without cup; cup filled to a flat surface at the top, i.e. no meniscus.							

Table 7.2: Lookup table of sample masses and resulting densities for density measurement cups included in delivery

7.4 Appendix 4: Order Information

Description	Order nr.
Liquid cups S4 40mm (500 pcs)	7KP19018FA
4µm Prolene foil, d=76mm (100 pcs)	BRE-CH3018
3.6µm Mylar foil, d=76mm (100 pcs)	BRE-CH3014
Sample cups 40mm prepared with 4µm Prolene (256 pcs)	RDL-1004-PRO
Sample cups 40mm prepared with 3.6µm Mylar (256 pcs)	RDL-1036-MY

Polystyrene disposable sample cups 3.5 mL	K410C30
Disposable pipettes (500 pcs)	7KP78048CC
Small balance max 100g; 0.01g display	K320C12

Table 7.3: Consumables for PETRO-QUANT 4.1

Description	**Order nr.**
Reference Sample Oil QC-XRS1	K410C277
Reference Sample Oil QC-XRS2	K410C279
Reference Sample Paraffin highly liquid	K410C281
Reference Sample Oil CONOSTAN S21&K&Sb -100 (QC)	K410C275
Reference Sample Oil CONOSTAN S21&K&Sb -500 (Drift)	K410C276

Table 7.4: Reference samples for PETRO-QUANT 4.1

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